

Granular Portfolios and the Origins of International Spillovers*

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Abstract

Capital-flow spillovers are traditionally understood through a “funding” channel, in which shocks to global banks propagate to the periphery. We identify a second origin: the granularity of country weights in portfolios. A delegated-portfolio model with sticky fund weights predicts that negative expectations about countries that account for large fund shares reduce capital flows to other countries held in the same funds, even though expectations about those countries are unchanged. This co-ownership spillover operates alongside the standard excess-return and hedging channels. We estimate each channel’s structural elasticity using a dataset that matches the GDP growth forecasts of global financial institutions to the assets under management and country allocations of their equity mutual funds. Co-ownership spillovers account for 60% of the variance of expectation-driven capital-flow reallocation, and a one-standard-deviation co-ownership shock moves country-level flows by between one-half and as much as a comparable shock to the VIX or the US interest-rate stance. Small advanced and emerging economies are the primary recipients; the G7 and BRICS are the main sources, producing a Large-to-Small rather than North-to-South transmission.

Keywords: International portfolio allocation; Capital flows; Expectations; Financial contagion; Portfolio frictions; Mutual funds.

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1 Introduction

International capital-flow spillovers are traditionally understood through a “funding” channel: shocks to global banks and major financial centers propagate to the periphery through internationally active banks (e.g., Schnabl, 2012; Rey, 2015; Miranda-Agrippino and Rey, 2020; Bruno and Shin, 2015; Aldasoro et al., 2023).¹ Yet intermediation has shifted toward non-bank financial intermediaries, and within them toward increasingly passive investment funds, now the dominant vehicle for cross-country equity allocation.² While these passive non-bank intermediaries transmit global funding shocks (e.g., Raddatz and Schmukler, 2012; Converse et al., 2023), they may trigger other forms of financial contagion.

This paper identifies a second origin of international financial spillovers: the granularity of country weights in sticky delegated portfolios. For example, suppose end-investors grow more optimistic about Brazil and channel more capital into a fund that holds Brazilian equity. Because the fund rebalances sluggishly, every country in its portfolio receives inflows, not Brazil alone, even though expectations about those countries remain unchanged. This “co-ownership spillover” channel³ transmits shocks *between* countries within a shared investment universe rather than from the investor base, and has been largely neglected in the literature thus far. Whether such spillovers generate meaningful capital-flow misallocation depends on the “granularity” of fund shares (Gabaix, 2011). If country-specific expectation shocks averaged out within portfolios, only global shocks (relevant for every country) would spill over. But these expectation shocks do not average out: fund portfolios are concentrated in a handful of large economies, so shocks specific to those countries survive aggregation and propagate to the rest of the cross-section through co-ownership. This gives international transmission a pattern different from the one the contagion literature has emphasized. Transmission does not run from North to South, but from Large to Small: the G7 and the BRICS are the main sources, while small advanced and emerging economies are the main recipients.

We formalize the mechanism and quantify its aggregate implications. We develop a delegated-portfolio model with sticky fund country weights and bring it to the data. The model decomposes expectation-driven capital flows into three channels: an *excess-return*

¹See the recent survey by Scheubel et al. (2025).

²For advanced-economy borrowers, portfolio debt rose above 50% of total external debt by 2019 while the share of cross-border bank loans fell from 34% to 20% (CGFS, 2021). Passive funds accounted for 48% of the assets under management of U.S. equity funds as of March 2020 (Anadu et al., 2020), and in Switzerland the share of domestic-currency fixed-income instruments held by mutual funds rose from 5% in 2005 to 51% in 2024 (Eugster et al., 2025). These portfolios are also geographically concentrated: the top 20 destination markets account for roughly 94% of external portfolio equity assets (Karolyi et al., 2020).

³The term is due to Jotikasthira et al. (2012).

channel through which country-specific expectations shift within-fund weights; a *co-ownership* channel through which fund-level expectations spill over mechanically across the fund’s holdings when portfolios are sticky; and a *hedging* adjustment driven by the covariance of country returns with the fund portfolio. The relative strength of the excess return and co-ownership channels depends on two structural elasticities: the elasticity of capital flows to the country-level expectation, and to the fund-level expectations. We identify these elasticities in the data and feed them back into the model to quantify each channel.

The key elasticities are identified using a unique dataset that matches the in-house country-level GDP growth forecasts produced by global financial institutions (typically large banks and asset managers) to the cross-country allocations of the equity mutual funds they manage. The matched panel covers 1,059 equity mutual funds managed by 54 financial institutions (among which are some of the world’s largest asset managers) investing across 49 countries over 2000-2023.⁴ Each institution is an information node: we take its in-house forecast as the empirical proxy for the information that shapes both its end-investors’ allocation across its funds and its managers’ allocation across countries within a fund.⁵ Mapping the model elasticities to a regression specification, we document a sharp asymmetry: inflows into a country within a given fund respond strongly to the fund’s in-house portfolio-weighted growth expectation, whereas they respond only weakly to country-level expectations. This asymmetry implies a substantial degree of portfolio stickiness, with a portfolio updating frequency of roughly eight months for the average fund (and six for active funds) when using our more conservative estimates.

Methodologically, our approach combines two complementary identification strategies. To address the missing-variable bias arising from incomplete observation of country-level expectations within fund portfolios, we construct a granular residual (Gabaix and Koijen, 2021, 2024) that aggregates the country-specific component of investor expectations using lagged portfolio weights.⁶ To address measurement error in subjective forecasts and potential reverse causality from capital flows to growth expectations, we further instrument the investor’s expectation with the contemporaneous IMF growth forecast, which is publicly available and is not directly affected by fund portfolios.

Plugging the elasticity estimates back into the three-channel decomposition, co-ownership spillovers account for 60% of the variance of expectation-driven capital-flow reallocation,

⁴Only 13 investors can be used in regressions that require fund-level expectations because a good portfolio coverage is required. However, these 13 investors are among the largest ones and manage 56% of the matched assets.

⁵We develop this institutional setting in Section 3.1 and assess the sensitivity of our estimates to this assumption with an IV strategy that uses publicly available IMF forecasts in Section 3.2.

⁶See the survey of papers using GIVs in Gabaix and Koijen (2024).

against 40% for the country-specific excess-return channel and less than 1% for hedging. This expectation-driven reallocation is economically large. In a horse race against the drivers of the global financial cycle, a one-standard-deviation co-ownership shock moves aggregate country-level capital flows by half to two-thirds as much as a comparable shock to the VIX, and by two-thirds to as much as a comparable shock to the US interest-rate stance. Because co-ownership spillovers are unrelated to recipient countries’ fundamentals, they constitute a source of cross-border capital misallocation that differs from the funding-shock channel traditionally emphasized in the contagion literature, yet of the same order of magnitude.

Through co-ownership spillovers, a small number of large economies generate substantial capital-flow reallocations across the broader cross-section of countries that share fund ownership with them.⁷ This produces a distinct pattern of contagion. Large advanced and emerging economies, such as the G7 and BRICS, are the main contributors, while small advanced and emerging economies are the primary recipients. As a result, some large emerging economies, such as China and Brazil, contribute strongly to spillovers while being relatively insulated themselves. These findings suggest that policymakers in small economies should pay attention not only to large shocks to global financial centers but also to shocks to large countries with overlapping investor bases.

What drives co-ownership spillovers is the *global structure* of portfolios, that is, the cross-country dispersion of *average* portfolio weights, and the corresponding concentration of global portfolios in a handful of countries: 10 countries account for 58% of portfolios in our sample. Compared to the granularity of country weights, the granularity of investors is second-order: the role of specific investors, when their portfolio allocation differs from the average, or when their expectations differ from the consensus, is negligible. As a result, *consensus* expectations drive the spillovers: they account for more than 90% of the variance of co-ownership spillovers. This also implies that our results generalize outside our dataset as long as its portfolio structure is representative.

Our first contribution is to the literature on granularity in financial markets. Existing work emphasizes that aggregate and cross-country fluctuations may arise from idiosyncratic shocks to large actors, such as firms or sectors, when size distributions are fat-tailed (Gabaix, 2011; Acemoglu et al., 2012; di Giovanni et al., 2014; Carvalho and Grassi, 2019; Herskovic et al., 2020; Gaubert and Itskhoki, 2021; Jamilov et al., 2026). In financial markets, related mechanisms operate through shocks to large banks, mutual funds, or other institutional investors (Ben-David et al., 2021; Amiti and Weinstein, 2018; Galaasen et al., 2020; Camanho et al., 2022; Gabaix and Koijen, 2021; Aldasoro et al., 2023; Bippus et al., 2023; Eugster et al., 2025). In contrast, granularity in our setting arises from the distribution of country

⁷Mutual funds manage a large and growing share of global capital flows; see Schmidt and Yesin (2022).

weights within portfolios, not from investor size. A small number of countries account for a disproportionate share of global equity fund holdings, so expectation shocks to these countries propagate internationally even when investors themselves are small and diversified.

Second, we contribute to the literature on international shock transmission through financial intermediaries, which shows that shocks to the investor base generate comovement across emerging markets and move asset prices (Broner et al., 2006; Coval and Stafford, 2007; Gelos, 2011; Raddatz and Schmukler, 2012; Jotikasthira et al., 2012; Puy, 2016), and to the closely related *common-ownership* literature in finance, which documents that assets held by the same institutions comove in excess of what fundamentals imply (Lou, 2012; Antón and Polk, 2014), including across borders when the common owners are foreign (Bartram et al., 2015). These findings are consistent with our mechanism being active in the data, though excess comovement admits other readings (e.g., Barberis et al., 2005). We differ in the nature of the shock. That literature takes fund flows as the primitive; we trace them to an observable economic source, the institution’s own growth expectations, which lets us separate the reallocation a country earns through its own prospects from the reallocation it inherits from its portfolio partners. The spillovers we identify are therefore neither funding shocks (e.g., Broner et al., 2006; Raddatz and Schmukler, 2012), nor rebalancing shocks (e.g., Camanho et al., 2022), nor shocks to specific granular institutions (e.g., Ben-David et al., 2021). In Jotikasthira et al. (2012), our closest antecedent, co-ownership spillovers are conjectured as a channel in a calibrated model of fund fire sales, but they are not the object of the paper’s empirical analysis. We take the co-ownership channel itself as our object of identification and quantification.

Third, we connect two literatures: survey-based expectations and portfolio adjustment frictions. In the first, a growing body of work links elicited beliefs to investment behavior.⁸ Closest to us, Dahlquist and Ibert (2024) show that asset managers tilt portfolios toward assets about which they are optimistic, and De Marco et al. (2022) and Li and Ongena (2025) show that the cross-country allocation of sovereign bond holdings and of bank credit supply responds to investors’ expectations.⁹ In the second, international portfolio flows are known to be persistent and only gradually responsive (Bohn and Tesar, 1996; Froot et al., 2001; Bacchetta et al., 2023), and funds’ country allocations track index weights rather than country fundamentals (Raddatz et al., 2017; Pandolfi and Williams, 2019; Pavlova and Siko-

⁸See Vissing-Jorgensen (2003); Glaser and Weber (2005); Kézdi and Willis (2011); Weber et al. (2012); Piazzesi and Schneider (2009); Malmendier and Nagel (2015); Agarwal et al. (2022); Giglio et al. (2021); Ma et al. (2022).

⁹In the demand-system approach to asset pricing, demand elasticities are inferred from holdings data, with beliefs left latent in the demand residual (Kojien and Yogo, 2019; Kojien et al., 2024; Kojien and Yogo, 2026; Fang et al., 2025). Because we observe the beliefs of the institution managing the fund, we can estimate elasticities with respect to beliefs directly.

rskaya, 2023). Our contribution is at the intersection: we estimate *two* belief elasticities—of fund flows and of within-fund country weights—whose ratio maps directly into the degree of portfolio stickiness. The corresponding portfolio updating frequency is higher than the estimates of Bacchetta and van Wincoop (2017) based on macroeconomic moments, but still low enough to generate sizeable misallocation. The novel result is that while portfolio shares are sticky, the flows into those portfolios are not: reallocation does occur, but it is *coarse*. It happens across blocks of countries rather than within them, and hence mostly across large countries rather than small ones. Our micro estimates are in this respect consistent with the macro evidence that expectation shocks account for the bulk of the variation in gross capital flows (Benhima and Cordonier, 2022).

Section 2 develops a delegated portfolio-choice model and characterizes co-ownership spillovers. Section 3 describes the data, maps the model to the data, and identifies the key elasticities. Section 4 quantifies the contribution of co-ownership spillovers. Section 5 concludes.

2 Model

This section develops a two-period delegated-portfolio model. The building blocks of the model are standard: mean-variance preferences and a Calvo-style portfolio adjustment friction (Bacchetta et al., 2022).¹⁰ The new element is their combination in a delegated setting. Namely, investors choose how to allocate their wealth across funds, and fund managers choose how to adjust the fund allocation across countries, but with a fixed probability that is less than one. When the fund portfolio weights do not adjust, they remain fixed at some default values. We assume that investors and fund managers share the same information and the same objective, so that the spillover mechanism we identify arises purely from the sticky fund portfolio adjustment.

We first lay out the model’s assumptions. Second, we decompose country-level capital flows into an excess-return channel, a co-ownership spillover, and hedging terms. We then aggregate these flows, show that the friction affects only portfolio reallocation, not portfolio growth, and pinpoint the key parameters to estimate.

¹⁰Bacchetta et al. (2022) use explicit adjustment costs and show that the modeling of the portfolio friction does not affect the main implications.

2.1 Setup

Agents. There are M investors indexed by $i = 1, \dots, M$. Each investor i is associated with $\mathcal{J}(i)$ equity mutual funds indexed by $j = 1, \dots, \mathcal{J}(i)$. There are K countries in which the funds can invest, indexed by $k = 1, \dots, K$, and fund (i, j) invests in a subset $\mathcal{S}(i, j)$ of $K(i, j)$ countries.

Returns and information. An equity share held in country $k = 1, \dots, K$ is traded at a normalized price of one in the first period and pays a stochastic return R_k in the second period. Stacking returns into the vector $R = (R_1, \dots, R_K)'$, we assume $R \sim \mathcal{N}(\bar{R}, \Sigma)$, with unconditional mean $\bar{R}$ and variance-covariance Σ .

Investor i and the funds it is associated with share the same information about R . Within the first period we distinguish beginning-of-period information $\bar{\mathcal{I}}^i$ from end-of-period information $\mathcal{I}^i$, with corresponding operators $\bar{E}^i(\cdot)$ and $E^i(\cdot)$, variances $\bar{V}(\cdot)$ and $V(\cdot)$, and shorthand $\bar{V}^R = \bar{V}(R)$ and $V^R = V(R)$. Throughout, we assume learning between the two stages is gradual:

Assumption 2.1 $\bar{V}^R - V^R \ll V^R$.

This assumption ensures that the change in the conditional variances between the two stages of the first period is small, which keeps the default portfolio weights tractable.

The investor's problem. Investor i enters the first period with wealth Ω^i , allocates a share $a^{i,j}$ to each fund j , and the residual $1 - \sum_j a^{i,j}$ to a safe asset paying r . The decision is taken after observing $\mathcal{I}^i$ but *before* fund j 's realized country allocation is known. Second-period terminal wealth is $[\mathcal{R}_p^i (\sum_j a^{i,j}) + r (1 - \sum_j a^{i,j})] \Omega^i$, where $\mathcal{R}_p^i$ is the return on the investor's portfolio of funds:

$$\mathcal{R}_p^i = \sum_{j=1}^{\mathcal{J}(i)} \frac{a^{i,j}}{\sum_{j'} a^{i,j'}} R_p^{i,j}, \quad (1)$$

with $R_p^{i,j}$ the period-2 return of fund j (defined below). The investor chooses $a^i = (a^{i,1}, \dots, a^{i,\mathcal{J}(i)})'$ to maximize a standard mean-variance objective with risk aversion γ , conditional on $\mathcal{I}^i$.

The fund's problem. The country allocation of fund (i, j) determines its portfolio return

$$R_p^{i,j} = \sum_{k \in \mathcal{S}(i,j)} w_k^{i,j} R_k = (w^{i,j})' R, \quad (2)$$

where $w_k^{i,j}$ is the share of fund j invested in country k . The portfolio friction operates at this stage. At the beginning of the period, fund weights are fixed at predetermined values $\bar{w}_k^{i,j}$ conditional on $\bar{\mathcal{I}}^i$. At the end of the period, once a^i has been chosen, fund j reoptimizes its weights with probability $p \leq 1$, and otherwise keeps them at $\bar{w}_k^{i,j}$. When reoptimizing, the fund maximizes the same mean-variance objective as the investor subject to $\sum_{k \in \mathcal{S}(i,j)} w_k^{i,j} = 1$, taking the distribution of country returns as given.

2.2 A three-channel decomposition of country flows

Conditional on the investor's end-of-period information $\mathcal{I}^i$ but *before* learning whether the fund will reoptimize, the expected share of investor i 's wealth channeled to country k through fund j , $a_k^{i,j}$ is the probability-weighted sum of the updated and default allocations:

$$a_k^{i,j} = p a_k^{i,j*} + (1 - p) \bar{w}_k^{i,j} a^{i,j}, \quad (3)$$

where $a_k^{i,j*}$ is the share of i 's wealth allocated to country k through fund j when fund j updates its portfolio, and $\bar{w}_k^{i,j} a^{i,j}$ is the share of wealth when fund j does not update its portfolio. Indeed, in the absence of updating, the funds deposited by investor i in fund j , $a^{i,j}$, are allocated to country k according to the default weight $\bar{w}_k^{i,j}$. Solving for $a_k^{i,j*}$ and $a^{i,j}$, and taking into account the fund's optimal setting of the default portfolio shares, yields a representation in which country-level flows respond to expectation surprises ("news") at three nested levels (country, fund, and the investor's overall portfolio):

Proposition 2.1 *Let $E^i(r_k) = E^i(R_k) - \bar{E}^i(R_k)$, $E^i(r_p^{i,j}) = E^i(R_p^{i,j}) - \bar{E}^i(R_p^{i,j})$, and $E^i(\mathbf{r}_p^i) = E^i(\mathcal{R}_p^i) - \bar{E}^i(\mathcal{R}_p^i)$ denote the period-2 "news" about expected country, fund, and investor portfolio returns. Under Assumption 2.1,*

$$\begin{aligned} a_k^{i,j} = & \underbrace{p \frac{E^i(r_k)}{\gamma V_k^{i,j}}}_{\text{Excess return}} + \underbrace{(1 - p) \frac{\bar{a}_k^{i,j}}{\bar{E}^i(a^{i,j})} \frac{E^i(r_p^{i,j})}{\gamma V^{i,j}}}_{\text{Co-ownership spillover}} - \underbrace{\Delta Cov_k^{i,j} \frac{E^i(r_p^{i,j})}{\gamma V^{i,j}}}_{\text{Hedging (fund-level)}} \\ & - \underbrace{\widetilde{Cov}_k^{i,j} \frac{E^i(\mathbf{r}_p^i)}{\gamma V^i} - (1 - p) \left(\frac{\bar{a}_k^{i,j}}{\bar{E}^i(a^{i,j})} Cov^{i,j} - Cov_k^{i,j} \right) \frac{E^i(\mathbf{r}_p^i)}{\gamma V^i}}_{\text{Hedging (investor-level)}} \\ & + \bar{E}(a_k^{i,j}), \end{aligned} \quad (4)$$

where $\Delta Cov_k^{i,j} = (Cov(R_k, R_{p,k-}^{i,j}) - Cov(R_k, \mathcal{R}_p^{i,j-})) / V_k^{i,j}$, with $R_{p,k-}^{i,j} = \sum_{k', k' \neq k} w_{k'}^{i,j} R_{k'} / (1 - w_k^{i,j})$ is the return of fund j portfolio when excluding country k and $\mathcal{R}_p^{i,j-} = \sum_{j', j' \neq j} a^{i,j'} R_p^{i,j'} / \left(\sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} a^{i,j'} \right)$ is the return of the full investor portfolio of funds when excluding fund

j . V^i , $V^{i,j}$, $Cov^{i,j}$, $Cov_k^{i,j}$, $\widetilde{Cov}_k^{i,j}$, $\bar{a}_k^{i,j}$, $\bar{E}^i(a_k^{i,j})$, and $\bar{E}^i(a^{i,j})$ are other constants defined in Appendix A.1.

Proof. See Appendix A.3. ■

Proposition 2.1 groups the response of country flows into three channels. The *excess-return* channel (first term) is the standard mean-variance response: a positive surprise in country k 's own expected return raises the allocation to k . However, the friction attenuates this elasticity: the channeling of more capital flows to country k can happen only if the fund reallocates its portfolio shares towards country k , which is less likely when $p < 1$. In the frictionless limit $p = 1$, this term reduces to the textbook mean-variance prediction.

The *co-ownership spillover* (second term) is the channel introduced by the delegated structure combined with sticky weights, and is the new mechanism of the paper. It loads on *fund-level* news $E^i(r_p^{i,j})$, which can be driven by expectations about any country in fund j 's portfolio, not just k . To see why, suppose news raises the expected return on some other country $k' \neq k$ inside fund j . The investor responds by increasing her allocation $a^{i,j}$ to the fund (Equation 30). In the frictionless benchmark, the fund manager would immediately rebalance towards k' alone, leaving k unaffected. With sticky weights, the fund instead splits the inflow across all its countries in proportion to its predetermined weights $\bar{w}_k^{i,j}$, mechanically channeling capital to k . The intensity of the spillover is governed by the share $\bar{a}_k^{i,j}/\bar{E}^i(a^{i,j})$, which is roughly the ex-ante fraction of fund j 's flows going to k . The spillover vanishes in the frictionless limit $p = 1$.

The remaining *hedging* terms collect mean-variance hedging adjustments, operating both at the fund and at the investor level. They reflect how country k co-varies with the rest of fund j 's portfolio and with the rest of the investor's portfolio. They are not the focus of the mechanism, but matter quantitatively for the identification of p below. However, it is important to note at this stage that the fund-wide hedging term, which loads on the news on fund j 's return $E^i(r_p^{i,j})$, is independent of the portfolio friction p . Indeed, the hedging of fund flows arise automatically from the optimal fixed part of the portfolio share $-\Delta Cov_k^{i,j}$, which does not depend on new information.¹¹ This fixed part $-\Delta Cov_k^{i,j}$ depends on the hedging properties of country k with regards to the rest of the fund portfolio (first covariance term) and to the investor's wider portfolio (second covariance term). Indeed, as $a^{i,j}$ rises, the investor's exposure to the fund rises, while the exposure to the rest of her portfolio falls; the fund therefore allocates less to k when k is a poor hedge for the fund and more when k is a

¹¹Here, Assumption 2.1 ensures that the "fixed" part of the portfolio shares, which depends on the ratio of the conditional covariance to the variance, is invariant whether the fund updates its shares or not and that the default shares are not significantly affected by any precautionary behavior. The investor-level hedging term, that loads on the news on the investor's overall portfolio return $E^i(r_p^i)$, depends on p in general, but becomes independent from p under the symmetry assumption we use for aggregation (Section 2.3).

poor hedge for the investor's wider portfolio.

Equation (4) can be re-expressed compactly in terms of three elasticities to expectation surprises:

Corollary 2.1 *Equation (4) can be rewritten as*

$$\frac{a_k^{i,j} - \bar{E}^i(a_k^{i,j})}{\bar{E}^i(a_k^{i,j})} = \beta_k^{i,j} E^i(r_k) + \delta_k^{i,j} E^i(r_p^{i,j}) + \theta_k^{i,j} E^i(\tau_p^i), \quad (5)$$

where $\beta_k^{i,j}$, $\delta_k^{i,j}$, and $\theta_k^{i,j}$ are the elasticities of capital flows to country-, fund-, and investor-level expectation surprises. The fund-level elasticity admits the further decomposition

$$\delta_k^{i,j} = \eta_k^{i,j} - \phi_k^{i,j} \Delta Cov_k^{i,j}, \quad (6)$$

where $\eta_k^{i,j} \propto 1 - p$ is the co-ownership elasticity and $\phi_k^{i,j} \Delta Cov_k^{i,j}$ is the fund-level hedging elasticity. Explicit expressions for $\beta_k^{i,j}$, $\eta_k^{i,j}$, $\phi_k^{i,j}$, and $\theta_k^{i,j}$ are given in Appendix A.3.

The two equations (5) and (6) are the structural representation that will map to regressions in the empirical section.

2.3 Aggregation and the role of the friction

We now aggregate the investor-fund flow surprises of Corollary 2.1 to the country level, and characterize how the portfolio friction shapes the aggregate response of capital flows to expectations.

From investor-fund flows to country flows. Total capital flows to country k are $A_k = \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} A_k^{i,j}$, where $A_k^{i,j} = a_k^{i,j} \Omega^i$, where Ω^i is total investor wealth. We focus on $a_k = A_k / \Omega$, the share of global wealth $\Omega = \sum_{i=1}^M \Omega^i$ flowing to country k :

$$a_k = \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\Omega^i}{\Omega} a_k^{i,j}. \quad (7)$$

Scaling by the ex-ante share, the surprise in country- k flows is a weighted average of investor-fund surprises:

$$\frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} = \sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \frac{a_k^{i,j} - \bar{E}^i(a_k^{i,j})}{\bar{E}^i(a_k^{i,j})}, \quad (8)$$

where $\sigma_k^{i,j} = \bar{E}^i(A_k^{i,j})/\bar{E}^i(A_k^i)$ is fund j 's ex-ante share in investor i 's flows to k , and $\sigma_k^i = \bar{E}^i(A_k^i)/\bar{E}(A_k)$ is investor i 's ex-ante share in aggregate flows to k , with $\bar{E}(A_k) = \sum_{m=1}^M \bar{E}^m(A_k^m)$. The investor-fund surprises receive more weight when the fund's average contribution to country k is larger.

Structure of expectations. According to Corollary 2.1, the capital-flow surprises depend on the return expectations. We impose a simple structure on these expectations:

Assumption 2.2 (Structure of expectations) *We assume that expectations $E^i(r_k)$ are equal to the sum of a global component W^i and an idiosyncratic country-specific one l_k^i :*

$$E^i(r_k) = W^i + l_k^i \quad (9)$$

with $E(l_k^i) = 0$, $Cov(l_k^i, W^i) = 0$ and $Cov(l_k^i, l_{k'}^i) = 0$ for all i and $k \neq k'$.

Under Assumption 2.2, the common component W^i can be recovered as the simple cross-country average of investor i 's expectations, and the local components l_k^i as the country-specific residual. Substituting into the fund- and investor-portfolio return expectations gives a decomposition of fund and investor return news into a granular term and a global term:

$$E^i(r_p^{i,j}) = \Gamma^{i,j} + W^i, \quad \Gamma^{i,j} \simeq \sum_{k \in \mathcal{S}(i,j)} w_k^{i,j} l_k^i = (w^{i,j})' l^i, \quad (10)$$

$$E^i(r_p^i) = \Gamma^i + W^i, \quad \Gamma^i \simeq \sum_{k=1}^K w_k^i l_k^i = (w^i)' l^i, \quad (11)$$

where $l^i = (l_1^i, \dots, l_K^i)'$ stacks the local components and $w_k^i = \sum_j a_k^{i,j} / \sum_j \sum_{k'} a_{k'}^{i,j}$ is country k 's share at the investor level. The granular components $\Gamma^{i,j}$ and Γ^i are weighted averages of the investor-specific country shocks l_k^i , with weights given respectively by fund-level and investor-level country shares. They will be key elements in the aggregation of capital flows and will play a key role for the co-ownership spillovers.

In order to aggregate capital flows, we define the aggregated version of the coefficients:

Definition 2.1 (Aggregate coefficients) *For $x \in \{\beta, \delta, \eta, \phi \Delta Cov, \theta\}$, define $x_k^i = \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} x_k^{i,j}$, $x_k = \sum_{i=1}^M \sigma_k^i x_k^i$ and $x = \sum_{k=1}^K \sigma_k x_k$, with $\sigma_k = \bar{E}(a_k)/\bar{E}(a)$ the ex-ante share of country k in total equity investments.*

The aggregation result also relies on two technical assumptions: an orthogonality condition (Assumption A.1) that lets us aggregate the elasticity coefficients, and a symmetry condition (Assumption A.2) that lets us treat them as homogeneous across countries within a fund. Both are stated and discussed in Appendix A.4.

Combining the investor-fund flow surprises in Corollary 2.1, the aggregation formula (8), and the Assumption on the structure of expectations 2.2, we obtain the central aggregate result:

Proposition 2.2 *Under Assumptions 2.2, A.1, and A.2 (see Appendix A.4 for the latter two), with $\Theta = \beta + \delta + \theta$ and $\sigma^i = \bar{E}^i(A^i)/\bar{E}(A)$ the ex-ante share of investor i in total equity, country- k flow surprises decompose as:*

$$\begin{aligned} \frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} = & \underbrace{\beta \left(\sum_{i=1}^M \sigma_k^i (l_k^i - \Gamma^i) \right) + \delta \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (\Gamma^{i,j} - \Gamma^i) \right)}_{\tilde{\Delta}a_k} \\ & + \underbrace{\Theta \left(\sum_{i=1}^M \sigma^i (W^i + \Gamma^i) \right)}_{\Delta a} \end{aligned} \quad (12)$$

where $\beta \propto p$ and $\delta = \eta - (\phi \Delta Cov)$ with $\eta \propto 1 - p$.

Proof. See proof in Appendix A.4. ■

Proposition 2.2 decomposes country- k flow surprises into a *portfolio reallocation* component $\tilde{\Delta}a_k$ and a *portfolio growth* component Δa , by rewriting $\frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)}$:

$$\frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} = \underbrace{\frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} - \frac{a - \bar{E}(a)}{\bar{E}(a)}}_{\tilde{\Delta}a_k} + \underbrace{\frac{a - \bar{E}(a)}{\bar{E}(a)}}_{\Delta a}, \quad (13)$$

where $a = A/\Omega$ and $A = \sum_{k=1}^K A_k$ is total equity and Ω is the total wealth of investors. Portfolio reallocation $\tilde{\Delta}a_k$ captures changes in country k 's share of equity flows (capital moving between countries). Portfolio growth Δa captures changes in total equity flows (capital moving into or out of equity overall). The proposition separates how each responds to expectations. Portfolio growth Δa is driven only by the investor-wide common components $W^i + \Gamma^i$, which raise the demand for equity across all countries and funds uniformly. Portfolio reallocation $\tilde{\Delta}a_k$ has two sources: $l_k^i - \Gamma^i$, the country-specific excess-return expectation relative to the rest of investor i 's portfolio (with elasticity β), and $\Gamma^{i,j} - \Gamma^i$, the fund-specific excess-return expectation relative to the rest of investor i 's portfolio (with elasticity δ). The first is the standard excess-return motive; the second includes the standard hedging motive and the co-ownership spillover.

Whether the friction matters for these objects is not obvious a priori. Reallocation depends on β and δ , both functions of p ; growth depends on the aggregate $\Theta = \beta + \delta + \theta$,

which mixes all three elasticities. The next corollary summarizes the comparative statics:

Corollary 2.2 *We assume that Assumptions A.1 and A.2 are satisfied. In that case,*

- (i) β is decreasing in $1 - p$,
- (ii) δ is increasing in $1 - p$ and is positive for a large $1 - p$,
- (iii) Θ is independent of p ,
- (iv) $\beta/\eta = p/(1 - p)$.

Proof. See proof in Appendix A.5. ■

The corollary provides the following insights. First, the country-expectation elasticity β is dampened by the friction: when the funds' country allocation is sticky, capital flows respond less to country- k -specific news than they would in the frictionless benchmark (point (i)). Second, the friction inflates the response to the granular component. Because $\eta \propto 1 - p$, as portfolios become stickier the co-ownership spillover grows; if $1 - p$ is large enough that η dominates the hedging adjustment $\phi\Delta Cov$, then δ turns positive and the granular term generates extra capital-flow volatility (point (ii)). Third, the friction does not affect the response to common components: Θ is the same as in the frictionless world (point (iii)). Together, these three results imply that the friction reshapes the cross-country *portfolio reallocation* of capital but leaves the aggregate *portfolio growth* of equity unchanged.

In the remainder of the paper, we thus focus on portfolio reallocation $\tilde{\Delta}a_k$, which can be decomposed into three terms:

$$\tilde{\Delta}a_k = \underbrace{\beta \tilde{l}_k}_{\text{excess return}} + \underbrace{\eta \Gamma_k}_{\text{co-ownership spillover}} - \underbrace{\phi \Delta Cov \Gamma_k}_{\text{hedging}}, \quad (14)$$

where $\tilde{l}_k$ is a country-specific excess-return aggregator and Γ_k is a granular aggregator:

$$\tilde{l}_k = \sum_{i=1}^M \sigma_k^i (l_k^i - \Gamma^i), \quad (15)$$

$$\Gamma_k = \sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (\Gamma^{i,j} - \Gamma^i). \quad (16)$$

We used the expression of $\tilde{\Delta}a_k$ in Proposition 2.2 and substituted $\delta = \eta - \phi\Delta Cov$ (Corollary 2.1). $\tilde{l}_k$ captures investors' *excess* optimism about country k relative to their overall portfolios, aggregated using investors' shares in flows to k σ_k^i . Γ_k captures investors' *excess*

optimism about funds investing in k relative to their overall portfolios, aggregated using fund- and investor-level flow shares σ_k^i and $\sigma_k^{i,j}$.

The friction increases the contribution of the granular term Γ_k to portfolio reallocation through $\eta \propto 1 - p$, while it dampens the contribution of the country-specific excess-return through $\beta \propto p$. β and η are thus key objects of interest that we will seek to identify.

Finally, as point (iv) of the corollary states, the ratio $\beta/\eta = p/(1 - p)$ provides a model-based mapping from the data to the friction parameter p . If the elasticities β and η can be identified empirically, then their ratio pins down the implied portfolio updating frequency. The next section turns to the identification and estimation of these parameters.

3 Identification and Estimation

Here we describe the matched panel of investor expectations and mutual-fund allocations. We then map Corollary 2.1 into a regression specification that identifies β (the excess-return channel), η (the co-ownership spillover), and ϕ (the hedging adjustment).

3.1 Data and institutional setting

Our dataset combines economic expectations data from Consensus Economics with mutual fund data from Emerging Portfolio Fund Research (EPFR).

Institutional setting. The financial institutions we study are typically global financial groups (global banks and asset managers) that conduct two activities relevant for our analysis. The first is asset management: the institution offers the possibility for end-investors (their individual and institutional clients) to place their assets in internally managed equity mutual funds, among other investment vehicles. The managers of these mutual funds allocate capital across countries on behalf of these end-investors. For instance, a wealth management officer at UBS advises her clients to place funds in UBS Latin America. The manager of UBS Latin America allocates these funds by either following a passive strategy (e.g. replicating the MSCI Latin America’s benchmark) if the mandate of the fund is passive, or by actively managing the fund’s country shares if it follows an active mandate. We obtain information on these funds through EPFR, which reports the funds’ assets under management (AUM) and fund flows at a weekly basis and the country allocation at a monthly basis for a subset of the funds.

The second activity is in-house macroeconomic research: a forecasting department produces country-level forecasts of GDP growth and other indicators, which the institution typ-

ically publishes internally. For instance, following the same example, UBS Macro Research publishes growth forecasts about individual Latin American countries that are propagated across UBS through internal communications, sales material, and investment committees, to which both the end-investor (either directly or through her UBS wealth management officer) and the fund manager are exposed. We obtain these forecasts through Consensus Economics, a firm that surveys professional forecasters on a monthly basis.

Throughout the paper, we refer to the financial institution as the “investor,” to its in-house forecast as the “investor’s expectation,” and to the mutual funds it manages as the “investor’s funds.” This maps into the model of Section 2: investor i is the representative end-investor advised by institution i , whose choice set is the set of funds that institution i manages, and fund j of investor i is a fund in the EPFR data matched to the institution that manages it. This terminology is based on the assumption that the institution’s in-house forecast is a reasonable proxy for the information used by both decision-makers, the end-investor allocating capital across funds and the fund manager allocating capital across countries within a fund.

Expectation dataset: Consensus Economics Data. We use forecast data from Consensus Economics, a survey firm that collects monthly projections from professional forecasters. Each month, respondents provide their current-year and next-year forecasts for key macroeconomic indicators for a range of countries. Our primary variable of interest is the forecasted real GDP growth for 51 advanced and emerging economies. Consensus Economics reports the institutional affiliation of each forecaster, which we extract, clean, and match to the corresponding financial institutions reporting information in the EPFR mutual fund data. Throughout, we use investor i ’s month- t next-year GDP growth forecast for country k , denoted $E_{t, g_{k,t}}^{i, \text{next year}}$, as the empirical proxy for the return $E_t^i(r_{k,t+1})$ defined in Section 2.¹²

By using in-house forecasts, we are relying on expectations that are a better reflection of the decision-makers’ subjective expectations, instead of the objective expectations (realized outcomes or statistical forecasts) traditionally used in the capital-flow literature to estimate investment elasticities (e.g., Bohn and Tesar, 1996; Curcuru et al., 2011; Koijen and Yogo, 2019). While this tradition assumes rational expectations, our approach allows for deviations from it, which is consistent with a now-substantial body of evidence rejecting rational expectations in macroeconomic forecasting (Bordalo et al., 2020; Adam et al., 2025). The distinction matters for portfolio decisions: Greenwood and Shleifer (2014) show that survey expectations of stock returns comove strongly with flows into equity mutual funds while

¹²Table D.1 in the Appendix shows that GDP growth and equity returns are positively correlated. GDP growth alone explains 10% of the yearly variation in equity returns.

being negatively correlated with objective, model-based expected returns, so that the two notions of expectations have opposite empirical implications for investment. Our matched data on the financial institutions’ *own* GDP growth forecasts let us sidestep this issue.

Investor and mutual fund dataset: EPFR Data. EPFR provides monthly fund-level country allocations, cash shares, assets under management, weekly fund flows, and valuation changes. These data are widely used to study international equity and bond investments and capture between 5 and 20% of market capitalization for most countries; existing work shows they closely match the Center for Research in Security Prices and balance-of-payments data in terms of equity flows and returns.¹³

The EPFR data identify the financial institution managing each fund. We match these institutions to those reported by Consensus Economics using token-based fuzzy matching complemented by manual verification using additional sources on corporate relationships.

Matched sample and cleaning. Our matched set of investors and mutual funds is a representative subset of the EPFR universe. As summarized in Table D.2, over our sample from January 2000 to December 2023, 3,346 mutual funds report monthly allocations in the EPFR data, with a median fund size of 338 million USD; the 1,059 funds we match to Consensus Economics are similar in size (median 334 million USD). Matching retains 54 of the 604 investors in the full EPFR allocation data, but these include some of the largest asset managers in the world (see Table D.3). Our OLS regressions keep fund-country pairs with forecast information and a fund allocation of at least 0.5%, yielding 168,000 observations across 827 funds, 50 investors, and 46 countries.

One limitation of the matched data is that we observe in-house expectations for only 24% of portfolios on average. We thus impose restrictions on the computation of fund-level expectations: we restrict the sample to countries with forecasts in at least 90% of the periods to limit country entry and exits into funds, and to funds investing in at least 6 such countries and with forecast coverage exceeding 20% of the portfolio. Fund-level expectations are eventually available for a smaller subset of the data that rest on a smaller number of 13 investors (see Tables D.2 and D.3), which nevertheless manage 56% of the assets under management of all matched investors.

3.2 Identification of β , η , and ϕ

We now add time subscripts to the model and map the structural equations to empirically testable regressions, at the country-allocation level and at the fund level. We assume that

¹³See Jotikasthira et al. (2012); Miao and Pant (2012); Schmidt and Yesin (2022).

the elasticities $\beta_k^{i,j}$, $\delta_k^{i,j}$, $\eta_k^{i,j}$, $\phi_k^{i,j}$, and $\theta_k^{i,j}$ defined in Corollary 2.1 are homogeneous across countries, investors, and funds, and we therefore drop the (i, j, k) indices. The identification strategy follows directly from Corollary 2.1.

Excess-return channel Equation (5) delivers a natural mapping to the data. Noting that $\frac{a_{k,t}^{i,j} - \bar{E}^i(a_{k,t}^{i,j})}{\bar{E}^i(a_{k,t}^{i,j})} \simeq \log(a_{k,t}^{i,j}) - \log(\bar{E}^i(a_{k,t}^{i,j}))$, and that $A_{k,t}^{i,j} = a_{k,t}^{i,j} \Omega_t^i$ Equation (5) can be rewritten as the following fund-country regression:

$$\log(A_{k,t}^{i,j}) = \beta E_t^i g_{k,t}^{\text{next year}} + \lambda_{k,t} + \lambda_t^{i,j} + \lambda_k^{i,j} + \epsilon_{k,t}^{i,j}, \quad (17)$$

where $A_{k,t}^{i,j}$ is fund j 's total allocation to country k at month t and $E_t^i g_{k,t}^{\text{next year}}$ is used as a proxy for $E_t^i(R_{k,t+1})$.

The share of investor assets allocated to fund j is $a_t^{i,j} = A_t^{i,j} / \Omega_t^i$. Because we include fund-time fixed effects $\lambda_t^{i,j}$, the specification is equivalent to one with $\ln(a_t^{i,j})$ as the dependent variable, with Ω_t^i absorbed in the fund-time fixed effects. The fund-time fixed effects then absorb the investor- and fund-level expectation surprises $\delta E_t^i(r_{p,t+1}^{i,j}) + \theta E_t^i(\tau_{p,t+1}^i)$ from Corollary 2.1. Fund-time fixed effects also capture other unobserved developments at the investor level (including funding shocks identified in the literature and alternative investment opportunities) that could be correlated with expectations. Country-investor-fund fixed effects $\lambda_k^{i,j}$ absorb the time-invariant terms $-\beta \bar{E}^i(R_{k,t+1}) + \log(\bar{E}^i(a_{k,t}^{i,j}))$, capturing time-invariant fund preferences across countries. This specification provides an estimate for β , the elasticity to the country-level expectation. This elasticity reflects the reallocation between countries by fund managers.

Country-time fixed effects $\lambda_{k,t}$ play an important identification role. First, they account for changes in allocations mechanically driven by country asset prices. Second, they account for country growth and monetary policy that simultaneously drive the country's supply of capital and expectations. Third, they also capture potential general-equilibrium effects that could bias the estimated elasticity downward. For instance, if all investors become optimistic about a country, capital flows into the country are mitigated by the equilibrium increase in the equity price. These global surges in optimism are absorbed by $\lambda_{k,t}$, so that the coefficient β identifies the impact of a change in expectations that is specific to investor i , and can be interpreted as a partial-equilibrium elasticity.

Column (1) of Table 1 reports the OLS estimate of (17). The response of mutual funds to investor forecasts is significant but small: when an investor anticipates a 1 percentage point rise in a country's growth forecast, investment in that country increases by about 2.2% in all the funds managed by that investor, so a country with an initial 10% share will benefit

Table 1: Fund-Country Allocations and Investor Expectations

	(1)	(2)	(3)	(4)	(5)	(6)
	$\log(A_{k,t}^{i,j})$	$\log(A_{k,t}^{i,j})$	$\log(A_{k,t}^{i,j})$	$\log(A_{k,t}^{i,j})$	$\log(A_{k,t}^{i,j})$ IV (boot SE)	$\log(A_{k,t}^{i,j})$ IV (boot SE)
VARIABLES						
$E_t^i(g_k^{\text{next year}})$	0.022*** (0.007)				0.051*** (0.012)	
$E_t^i(g_p^{j,\text{next year}})$		0.174** (0.075)				
$\Gamma_t^{i,j}$			0.271** (0.120)	0.283** (0.136)		0.345*** (0.079)
$\Delta Cov_t^{i,j} \times \Gamma_t^{i,j}$				-0.173** (0.073)		-0.177** (0.077)
$\Delta \log(Q_t^{i,j})$		-0.011*** (0.003)	-0.011*** (0.003)	-0.011*** (0.003)		-0.010*** (0.001)
$\Delta \log(Q_{t-1}^{i,j})$		-0.007** (0.003)	-0.008*** (0.003)	-0.008** (0.003)		-0.007*** (0.001)
$\Delta \log(Q_{k,t})$					0.584*** (0.021)	
$\Delta \log(Q_{k,t-1})$					0.559*** (0.020)	
Observations	150,039	35,287	35,287	30,928	486,892	94,410
R-squared	0.974	0.926	0.926	0.926	0.943	0.904
Country-fund FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-time FE	Yes	No	No	No	No	No
Fund-time FE	Yes	No	No	No	Yes	No
Inv-country-time FE	No	Yes	Yes	Yes	No	Yes

Notes: Standard errors in parentheses. Columns (1)-(4) are OLS with analytic standard errors clustered at the investor-country level. Columns (5)-(6) report two-step IV estimates that use generated regressors from first-stage projections; their standard errors are computed via a pairs cluster bootstrap with 200 replications and clusters at the investor-country level, which propagates the first-stage estimation uncertainty into the reported standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

from a 0.22 percentage point increase. Table D.4 in the Appendix breaks the OLS results out by fund type: passive funds show no reallocation, while active funds react slightly more strongly ($\beta = 0.028$ in Columns (1) and (2)).¹⁴

Co-ownership and hedging channels We next focus on the elasticity to fund-level expectations, which captures the co-ownership and hedging channels jointly. Equation (5) can be rewritten as:

$$\ln(A_{k,t}^{i,j}) = \eta E_t^i g_{p,t}^{j, \text{next year}} - \phi \Delta Cov_k^{i,j} E_t^i g_{p,t}^{j, \text{next year}} + \lambda_k^j + \lambda_{k,t}^i + \epsilon_t^{i,j}, \quad (18)$$

where the portfolio expectation $E_t^i g_{p,t}^{j, \text{next year}}$ proxies for fund-level return news $E_t^i(r_{p,t+1}^{i,j})$. The specification separately identifies the two elasticities: η as the coefficient on the linear term and $-\phi$ as the coefficient on its interaction with $\Delta Cov_k^{i,j}$. Cross-fund variation in $\Delta Cov_k^{i,j}$ thus disentangles co-ownership from hedging, and the combined elasticity $\delta = \eta - \phi \Delta Cov$ can be read off as the coefficient on the linear term in a specification that omits the interaction. Appendix B.1 details the construction of $\Delta Cov_k^{i,j}$, and Appendix B.2 reports its summary statistics.

The fixed effects absorb a rich set of confounders. Country-investor-time fixed effects $\lambda_{k,t}^i$ absorb the country- and investor-level surprises $\beta E_t^i(r_{k,t+1}) + \theta E_t^i(\mathbf{r}_{p,t+1}^i)$ from Corollary 2.1, variations in the investor's total wealth $\log(\Omega_t^i)$, investor-specific outside investment opportunities and funding shocks, and the country-specific developments that were previously captured by $\lambda_{k,t}$. Country-investor-fund fixed effects $\lambda_k^{i,j}$ absorb the time-invariant terms $-\delta \bar{E}^i(R_{p,t+1}^{i,j}) + \log(\bar{E}^i(a_{k,t}^{i,j}))$.

We construct the portfolio expectation as a weighted average of country-level forecasts using the previous period's portfolio weights:

$$E_t^i g_{p,t}^{j, \text{next year}} = \sum_{k \in \mathcal{S}(i,j)} w_{k,t-1}^{i,j} E_t^i g_{k,t}^{j, \text{next year}}, \quad (19)$$

where $\mathcal{S}(i,j)$ is the set of countries in which fund j invests *and for which we observe investor i 's expectations*, so the weights $w_{k,t-1}^{i,j}$ do not necessarily sum to one. We return to this missing-expectation issue below.

A potential concern is that the documented fund-level elasticity could reflect the well-known *flow-performance* relationship in mutual funds: high past returns mechanically generate inflows (Chevalier and Ellison, 1997; Sirri and Tufano, 1998), and managers whose in-house forecasts have been more optimistic are typically those whose funds have recently

¹⁴A fund is classified as passive if its country allocation tracks a buy-and-hold benchmark.

performed well, so the apparent expectation-flow link could be spurious. To address this concern, we add to Equation (18) a measure of the fund’s portfolio return $\Delta \log(Q_t^{i,j})$, derived from the underlying asset prices in the portfolio, together with its lag.¹⁵ The estimated expectation elasticity reported below is therefore identified holding contemporaneous and recent past fund performance fixed.

Column (2) of Table 1 reports the results for Equation (18), without the interaction term. In that case, the estimated impact of the fund portfolio corresponds to the δ elasticity, that combines the co-ownership and hedging channels. The investor’s portfolio expectations are positively associated with the flows allocated to country k in the portfolio: an increase in the expected portfolio GDP growth by one percentage point is associated with an increase in investor allocations of about 17 percent.

Missing-expectation issue and granular instrument The coverage-weighted proxy in Equation (19) sums only over countries with observed forecasts, $\mathcal{S}(i, j)$. The true fund-level expectation also loads on the unobserved set $\tilde{\mathcal{S}}(i, j)$, generating a missing-variable bias that is not fully absorbed by the fixed effects.

To address this, we exploit the structure of expectations from Assumption 2.2, $E_t^i g_{k,t}^{\text{next year}} \simeq W_t^i + l_{k,t}^i$, and adopt a granular-instrumental-variable strategy à la Gabaix and Koijen (2024). The simple cross-country average over the observed set, $\frac{1}{K(i,j)} \sum_{k \in \mathcal{S}(i,j)} E_t^i g_{k,t}^{\text{next year}} \simeq W_t^i$, estimates the investor-common component, and the empirical counterpart of the granular term $\Gamma_t^{i,j}$ in Equation (10) is

$$\Gamma_t^{i,j} = \sum_{k \in \mathcal{S}(i,j)} w_{k,t-1}^{i,j} \left[E_t^i g_{k,t}^{\text{next year}} - \frac{1}{K(i,j)} \sum_{k \in \mathcal{S}(i,j)} E_t^i g_{k,t}^{\text{next year}} \right] \simeq \sum_{k \in \mathcal{S}(i,j)} w_{k,t-1}^{i,j} l_{k,t}^i, \quad (20)$$

where $K(i, j)$ is the number of countries in $\mathcal{S}(i, j)$. Since $\text{Cov}(l_{k,t}^i, W_t^i) = 0$ and $\text{Cov}(l_{k,t}^i, l_{k',t}^i) = 0$ for $k \neq k'$ by Assumption 2.2, $\Gamma_t^{i,j}$ is orthogonal to expectations on the missing countries in $\tilde{\mathcal{S}}(i, j)$, and the missing-variable bias is corrected.

Column (3) reports the regression in which the granular residual $\Gamma_t^{i,j}$ replaces $E_t^i g_{p,t}^{j, \text{next year}}$. The coefficient is positive and significant at 0.27. Column (4) adds the interaction term, which disentangles the co-ownership channel from the hedging channel. The interaction coefficient is significantly negative at -0.17 , consistent with the model and implying $\phi = 0.17$: hedging is at play. Investors consider the covariance of returns and the potential for risk-sharing (when ΔCov is negative), as well as arbitrage opportunities (when ΔCov is positive).

¹⁵The results do not change if we construct $\Delta \log(Q_t^{i,j})$ based on country-level MSCI indices, or if we include additional lags to capture the slower retail-investor response documented in the flow-performance literature.

The coefficient of the linear term is 0.28, implying $\eta = 0.28$, not very different from the estimated δ . This is consistent with values of $\Delta Cov_k^{i,j}$ that is symmetrically distributed across positive and negative values, with a low average of 0.05 (see Appendix B.2).

IV estimations We have so far made the assumption that the financial institutions’ forecasts are a good representation of the decision-makers’ (the end-investors’ and the fund managers’) information. We have also made the assumption that the fund portfolios’ exposure to different countries does not influence these forecasts. Violation of these assumptions produce, respectively, a measurement error and a reverse causality issue.

We address both issues by instrumenting the in-house forecast with the contemporaneous IMF growth forecast. Being publicly available, the IMF forecast is a public signal that reaches forecaster and decision-maker alike, thus addressing measurement error.¹⁶ It is not shaped by any individual fund’s country exposures, thus addressing reverse causality. It also has much broader coverage: an IMF forecast exists for most portfolio countries even when the subjective forecast is missing, so the fund-level expectation can be computed over a larger share of each portfolio.¹⁷ We therefore run the first stage on the reduced sample of funds with subjective expectations and the second stage on the extended IMF-based sample, which substantially increases the number of usable observations (see Table D.2).

Columns (5) and (6) of Table 1 report the IV estimates. In Column (5), we estimate β by instrumenting $E_t^i(g_{k,t}^{\text{next year}})$ with $E_t^{IMF}(g_{k,t}^{\text{next year}})$. Because the IMF forecast is country- and time-specific but not investor-specific, we cannot include country-time fixed effects; we instead control for valuation effects with the contemporaneous and once-lagged change in the country’s log MSCI equity price $\Delta \log(Q_{k,t})$.¹⁸ Correcting for measurement error and reverse causality raises the elasticity to $\beta = 0.051$ (and to 0.067 for active funds, while passive funds remain inelastic; see Table D.4).

In Column (6), we estimate η by instrumenting the fund-level expectation $E_t^i(g_{p,t}^{j,\text{next year}})$ with its IMF counterpart, computed on the subjective-expectation universe in the first stage and on the full IMF universe in the second. The elasticity rises only slightly, to $\eta = 0.34$. Its stability across these two samples indicates both that the granular residual already handles the missing-expectation problem well and that the results survive tripling the sample; the modest increase suggests the in-house forecasts proxy the end-investor’s information well.

¹⁶See Appendix C for a detailed formalization of this measurement error issue.

¹⁷The IMF expectation data covers on average 88% of the funds’ portfolios against 24% for subjective expectations.

¹⁸Equity prices capture potential general-equilibrium and valuation effects, but they do not capture common local drivers of the supply of capital and growth expectations that were previously accounted for through country-time fixed effects. However, these effects, if they are at play, generate a positive bias in the estimated β , which will reduce the impact of co-ownership spillovers. Therefore, it is a conservative estimate.

Implied portfolio-updating frequency The estimated β and η provide a direct estimate of the portfolio friction. By point (iv) of Corollary 2.2, the portfolio adjustment probability satisfies $p = \beta/(\beta + \eta)$. Our IV estimates imply $p = 0.051/(0.051 + 0.34) \simeq 0.13$, corresponding to portfolio updating every 8 months for the average fund; for active funds, $p \simeq 0.17$, or once every 6 months. These updating probabilities are higher than the value $p = 0.022/(0.022 + 0.28) = 0.07$, or once every 15 months, implied by our OLS estimates, which, notably, coincides with the Bacchetta and van Wincoop (2017) estimate obtained through global equity portfolio shares and returns. We take the IV estimates as our baseline: they correct for measurement error and reverse causality, and they are also the more conservative choice, since a higher p implies less stickiness and a smaller role for co-ownership spillovers.

Taken together, the estimates validate the model’s mechanism: flows into funds respond strongly to investors’ fund-level expectations, while cross-country allocations within funds respond only weakly to country-specific expectations. With (β, η, ϕ) now identified, the remaining quantitative question is how much of expectation-driven capital-flow reallocation is accounted for by each of the three channels in Equation (14). The next section answers this.

3.3 Robustness

Some concerns might still be raised about our results: that the IMF-forecast IV breaks down once forecasters and decision-makers process information differently, that the estimates are driven by one or two large investors who dominate the matched panel, that they are sensitive to the sample-construction thresholds adopted in Section 3.1, and that they are sensitive to timing assumptions and to the forecast horizon. We address each in turn.

Alternative assumptions about expectation formation Our IV rests on forecasters and decision-makers weighting public information the same way. They may not: a decision-maker who views her private information as more precise (or who faces strategic concerns the forecaster does not) will put a different weight on the public IMF signal than the forecaster (e.g., Broer and Kohlhas, 2024; Adam et al., 2025; Ottaviani and Sørensen, 2006; Gemmi and Valchev, 2025). Such asymmetries bias the IMF-forecast IV. We show in Appendix C that the structural elasticity is then bracketed by a complementary pair of estimators: the IMF-forecast IV controlling for realized (vintage) GDP growth, and the vintage-growth IV controlling for the IMF forecast. Each isolates the forecaster’s weight on a single signal, public or private, so the two estimators bound the elasticity from opposite sides, whichever way the asymmetry runs.

Figure E.1 in the Appendix reports the resulting bounds for β and η . The two estimates are statistically indistinguishable, and both confidence intervals contain our baseline IMF-IV estimate, indicating that the forecaster/decision-maker asymmetry is empirically small. Even at the extremes, the upper bound for β (0.075) and the lower bound for η (0.35) imply an upper bound for p of 0.18, close to our baseline of 0.13; we use these values in one of the quantitative robustness exercises.

The influence of individual investors Having established that the IV is robust to forecaster/decision-maker asymmetries, we now turn to a different concern: that the estimates are driven by the panel’s most active institutions. Our panel is imbalanced, with a handful of large investors accounting for a disproportionate share of observations (see Table D.3). We re-estimate the baseline specifications removing one investor at a time, focusing on the 10 investors with the highest observation share. Figure E.2 in the Appendix shows that the leave-one-out estimates are stable and broadly in line with the full-sample results.¹⁹

Alternative data cuts Another concern is that the estimates depend on the sample-construction choices made in Section 3.1. Our baseline focuses on fund-country pairs with average fund allocations above 0.5%; we now also consider the cases with no filtering and with thresholds at 1% and 3%. Our baseline also computes fund-level expectations and their granular residuals for funds whose expectation coverage exceeds 20% of the portfolio and covers at least 6 countries; we now also consider versions with at least 10 countries and with coverage of at least 10%, 30% or 50%. Figure E.3 in the Appendix reports the results: the elasticities are similar across all specifications.

Timing and horizon The remaining issues concern our timing assumptions and shifting horizons. First, in the baseline, we assume that the expectations are observed and reacted on during the Consensus Economics publication month. In practice, Consensus Economics surveys professional forecasters at the beginning of the month, and then releases the survey in the course of the third week of the month. This suggests that the forecasts are produced in the first half of the month. We however cannot rule out that it takes longer to the internal releases to be advertised within the organization. We thus relax the contemporaneity assumption by using the expectations lagged by one month. The results are represented in Figure E.4. The estimates are very close to the baseline.

¹⁹One exception is the OLS estimate of the β elasticity obtained without the largest investor, which is non-significant and significantly different from the baseline. However, the IV estimate is in line with the baseline. It is likely that the forecasts of this particular investor differ more significantly from the fund managers’, generating a more severe measurement error.

Second, note that the use of monthly releases of next-year GDP growth generates a shifting horizon over the year: the forecast corresponds to a longer horizon in the early months of the year and to a shorter horizon in the later months. Our estimates potentially average out unequal reactions to different horizons. This could be concerning if the later months' reaction was more significant and stronger but not showing up in our estimates. To check this, we split the sample into the first 6 months and the last 6 months of the year. Figure E.4 shows that the results are not strongly affected by the shifting horizon.

4 Quantifying Co-ownership Spillovers

With (β, η, ϕ) identified, we now ask how much of expectation-driven capital-flow reallocation is generated by each of the three channels in Equation (14) (excess return, co-ownership spillovers and hedging). We focus on portfolio reallocation rather than portfolio growth because Corollary 2.2 implies that portfolio growth is not affected by p , while portfolio reallocation is. We then dissect co-ownership spillovers further: we examine in more details the recipients and sources of co-ownership spillovers, we determine which source of granularity matters (country-specific or investor-specific) and benchmark the magnitude of co-ownership spillovers against major drivers of funding shocks (the VIX and US monetary policy).

4.1 Decomposing capital flow reallocation

The three-channel decomposition in Equation (14) applies period by period, with $\tilde{l}_{k,t}$ and $\Gamma_{k,t}$ the time- t analogs of (15)-(16), constructed from contemporaneous weights $\sigma_{k,t}^i$, $\sigma_{k,t}^{i,j}$ and time- t expectation residuals $l_{k,t}^i$, Γ_t^i , $\Gamma_t^{i,j}$. We parametrize β , η , ϕ , and ΔCov using our estimates from Section 3, and compute $\Gamma_{k,t}$ and $\tilde{l}_{k,t}$ from fund allocations and investor GDP growth expectations.

Measurement and calibration The aggregators $\tilde{l}_{k,t}$ and $\Gamma_{k,t}$ measure investors' *excess* optimism about a country, and about the funds exposed to it, relative to the rest of the portfolio, aggregated with the flow shares $\sigma_{k,t}^i$ and $\sigma_{k,t}^{i,j}$ (Equations (15)-(16)). Their building blocks are the investor's country-specific expectations $l_{k,t}^i$, and the investor-, and fund-level

granular residuals $\Gamma_t^{i,j}$ and Γ_t^i , which we construct as

$$\begin{aligned} l_{k,t}^i &= E_t^i g_{k,t}^{\text{next year}} - \frac{1}{K} \sum_{k=1}^K E_t^i g_{k,t}^{\text{next year}}, \\ \Gamma_t^i &= \sum_{k=1}^K w_{k,t}^i \left(E_t^i g_{k,t}^{\text{next year}} - \frac{1}{K} \sum_{k=1}^K E_t^i g_{k,t}^{\text{next year}} \right), \\ \Gamma_t^{i,j} &= \sum_{k \in \mathcal{S}(i,j)} w_{k,t}^{i,j} \left(E_t^i g_{k,t}^{\text{next year}} - \frac{1}{K(i,j)} \sum_{k \in \mathcal{S}(i,j)} E_t^i g_{k,t}^{\text{next year}} \right), \end{aligned}$$

Because expectations are missing for many investor-country pairs, we expand coverage by imputing missing expectations using an estimated expectation process. The imputation procedure is described in details in Appendix B.3. The procedure takes the country consensus (the average across investors) as given, models the country-investor-specific component, and attributes fictitious expectation data to each investor with missing expectations. The expanded panel can thus exploit fully the EPFR allocation data, with the following cleaning: we use only fund-date units with at least 6 countries to compute fund-level averages (as in our empirical sample), and only country-date units with at least 3 forecasts to compute consensus (average) forecasts.²⁰ The clean sample then contains 549 investors, 2,921 funds, and 2,808,970 monthly observations (see Table D.2 in the Appendix).²¹

We estimate the flow-share weights as $\sigma_{k,t}^i = A_{k,t}^i / A_{k,t}$ and $\sigma_{k,t}^{i,j} = A_{k,t}^{i,j} / A_{k,t}^i$, then compute $\tilde{l}_{k,t}$ and $\Gamma_{k,t}$ from (15) and (16) using contemporaneous weights $\sigma_{k,t}^i$, $\sigma_{k,t}^{i,j}$ and $w_{k,t}^{i,j}$ (results are similar with lagged or average weights).

We set $\beta = 0.051$, $\eta = 0.34$, and $\phi = 0.17$ from our IV point estimates in Section 3, and $\Delta Cov = 0.07$, the average of $\Delta Cov_k^{i,j}$ across all fund-country pairs (Table B.1 in Appendix B.2).

Contributions Table 2 reports the variance of expectations and the contribution of each of the three channels in (14) (excess returns $\beta \tilde{l}_{k,t}$, co-ownership spillovers $\eta \Gamma_{k,t}$, and hedging $-\phi \Delta Cov \Gamma_{k,t}$) to the variance of expectation-driven capital flow reallocation $\tilde{\Delta} a_{k,t}$, with

²⁰We choose a low minimum of 3 expectations not to lose too many observations. We impose a more restrictive threshold in our robustness analysis.

²¹Because there are some countries in which investor i invests and for which we do not have expectations (real or imputed), we use the formulas $\Gamma_t^{i,j} = \sum_{k \in \kappa(i,j)} w_{k,t}^{i,j} \left[E_t^i g_{k,t}^{\text{next year}} - \sum_{k \in \kappa(i,j)} E_t^i g_{k,t}^{\text{next year}} / \kappa^{i,j} \right]$, and $\Gamma_t^i = \sum_{k \in \kappa(i)} w_{k,t}^i \left[E_t^i g_{k,t}^{\text{next year}} - \sum_{k \in \kappa(i)} E_t^i g_{k,t}^{\text{next year}} / \kappa^i \right]$, where $\kappa(i)$ and $\kappa(i,j)$ are the sets of countries for which we observe investor i 's expectations or impute expectations at the investor and fund level, and κ^i and $\kappa^{i,j}$ are their respective cardinalities. This amounts to setting the expectations of these countries to zero, so the estimated co-ownership spillovers will be under-estimated. In this sense, we provide a conservative estimate of the variance of $\Gamma_{k,t}$.

Table 2: Variance decomposition of expectation-driven capital flow reallocation

Expectations			
Variance	$V(\tilde{l}_{k,t})$	$V(\Gamma_{k,t})$	
Value	.78	.05	
	[.17,1.58]	[.02,.10]	
Implied capital flows			
Coefficients	β	η	$-\phi\Delta Cov$
	.051	.34	-.01
Variance	$V(\beta\tilde{l}_{k,t})$	$V(\eta\Gamma_{k,t})$	$V(-\phi\Delta Cov_{k,t})$
Value	.0020	.0063	.0000
	[.0004,.0041]	[.0018,.0117]	[.0000,.0000]
Contribution	24%	76%	.0%
	[11%,43%]	[57%,89%]	[.0%,.1%]
Variance	$V(\beta\tilde{l}_{k,t} + \eta\Gamma_{k,k,t})$	$V(\eta\Gamma_{k,t} - \eta\Gamma_{k,k,t})$	$V(-\phi\Delta Cov_{k,t})$
Value	.0039	.0054	.0000
	[.0007,.0084]	[.0013,.0108]	[.0000,.0000]
Contribution	40%	60%	.0%
	[14%,75%]	[25%,85%]	[.0%,.1%]

Notes: We report the average variances of expectations and implied capital flow reallocation across countries, as well as the 10th and 90th percentile (in brackets). The contributions are the ratio of the variance to the total variance of expectation-driven flow reallocation.

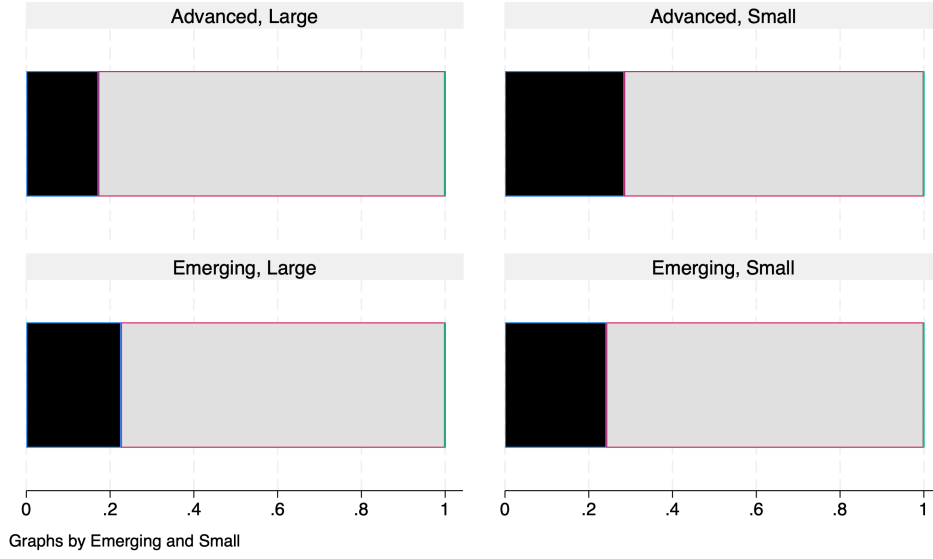
cross-country averages and $[p_{10}, p_{90}]$ ranges in brackets. Consider first the variance of expectations (upper part of the table): the idiosyncratic, country-specific term $\tilde{l}_{k,t}$ (0.78) is about 14 times as volatile, on average, as the granular term $\Gamma_{k,t}$ (0.05). Despite this, co-ownership (lower part of the table) contributes three times as much to the variance of reallocation as the excess-return channel (76% against 24%), because β is far smaller than η . The hedging component is negligible, at less than 1%.

Figure 1 splits countries by development status (Advanced/Emerging) and portfolio size (Large/Small). A “Large” country has an average share in portfolios in the top quartile (i.e., higher than 3.5%). The Large countries include the United States, the United Kingdom, Japan, Germany, France, Switzerland, the Russian Federation, South Korea, China, India, Mexico, and Brazil. The excess-return contribution is slightly higher for small economies (about 26%) than for large economies (about 20%).

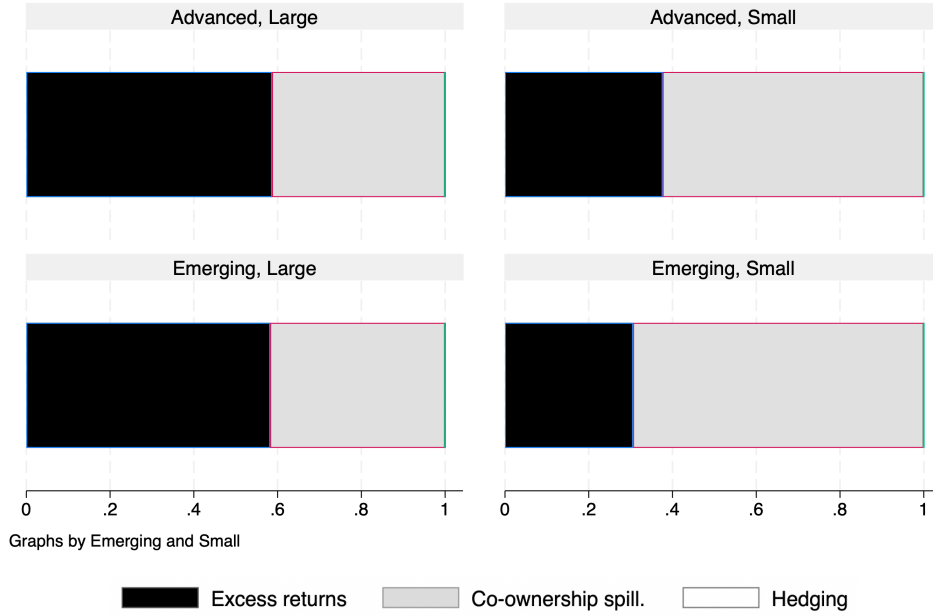
The granular aggregator $\Gamma_{k,t}$ for country k also loads on investors’ expectations about k itself, because k appears in the portfolios over which the aggregator is computed. For a large country, this own-country loading is non-trivial, and part of what we labelled a “co-

Figure 1: Variance decomposition of expectation-driven capital flows

a) Implied capital flows



b) Implied capital flows (adjusted)



Notes: Panel a) represents the contribution of the variance of $\beta \tilde{l}_{k,t}$, $\eta \Gamma_{k,t}$ and $-\phi \Delta Cov_{k,t}$ to the variance of implied capital flow reallocation. Panel b) represents the contribution of the variance of $\beta \tilde{l}_{k,t} + \eta \Gamma_{k,k,t}$, $\eta \Gamma_{k,t} - \eta \Gamma_{k,k,t}$ and $-\phi \Delta Cov_{k,t}$ to the variance of implied capital flow reallocation.

ownership spillover” is really the country’s own excess return acting through the granular term. To be conservative about how much of the reallocation we attribute to spillovers, we therefore isolate the own-country part

$$\Gamma_{k,k,t} = \sum_{i=1}^M \sigma_{k,t}^i w_{k,t}^i (l_{k,t}^i - \Gamma_t^i) \quad (21)$$

and reclassify $\eta \Gamma_{k,k,t}$ as part of the excess-return channel rather than the spillover channel. This yields a diminished co-ownership term $\eta(\Gamma_{k,t} - \Gamma_{k,k,t})$ and an augmented excess-return term $\beta \tilde{l}_{k,t} + \eta \Gamma_{k,k,t}$. Under this adjustment, the average co-ownership contribution falls from 76% to 60% and the excess-return contribution rises from 24% to 40%, with the change concentrated among large countries: in Panel b) of Figure 1, the co-ownership share among Large economies drops from 80% to 42%, while for Small countries it edges down only from 74% to 66%.

Robustness We test the sensitivity of the co-ownership contribution along three dimensions: the calibration of the friction parameter p , the measurement of portfolio weights, and the imputation of missing expectations. Table D.5 in the Appendix collects the results.

Consider first the calibration. Our benchmark uses $\beta = 0.051$ and $\eta = 0.34$, i.e. $p = 0.13$; since $\beta/\eta = p/(1-p)$ and the hedging term is negligible, the contribution depends effectively only on p . We bracket the benchmark with two polar values: the upper bound $p = 0.18$ identified in Section 3.3, and the lower estimate $p = 0.07$ taken from Bacchetta and van Wincoop (2017). The contribution remains large under the conservative calibration, at 46%, and rises to 83% under the value taken from the literature.

Turning to measurement, because the granular term Γ_k is built from the actual portfolio shares $w_{k,t}^{i,j}$, its variance could be inflated by the variance of those shares rather than by the volatility of expectations alone. Fixing the shares at their sample averages leaves the contribution essentially unchanged, at 58%.

Consider now the imputation of missing expectations. Raising the minimum number of forecasts per country-date from 3 to 6, so that country consensus (average) forecasts rely on more observations, yields 57%. Replacing the imputed investor-specific expectations $l_{k,t}^i$ with the cross-investor consensus $l_{k,t}$ gives 62%. In every case the co-ownership contribution stays substantial and is often close to the baseline, confirming that it is driven neither by the calibration, the weight dynamics, nor the imputation step.

4.2 Dissecting co-ownership spillovers

We proceed in two steps: we first express $\Gamma_{k,t}$ in terms of *co-ownership linkages* across countries, and then show that these linkages make the spillover operate through country-level granularity, rather than investor-level granularity.

The role of co-ownership linkages From the definitions of $\Gamma_t^{i,j}$ and Γ_t^i , the fund-level excess term $\Gamma_t^{i,j} - \Gamma_t^i$ is a weighted average of country-specific expectations:

$$\Gamma_t^{i,j} - \Gamma_t^i = \sum_{k=1}^K \Delta w_{k,t}^{i,j} l_{k,t}^i$$

where the weight $\Delta w_{k,t}^{i,j} = w_{k,t}^{i,j} - w_{k,t}^i$ is a *relative allocation*: the difference between country k 's share in fund j and its share in investor i 's full portfolio. The term matters to the extent that the fund's portfolio is concentrated relative to the investor's.

By replacing these terms in equation (16), the granular term for country k , $\Gamma_{k,t}$, can in turn be written as a weighted average of all country-specific expectations:

$$\Gamma_{k,t} = \sum_{k'=1}^K \sum_{i=1}^M \sigma_k^i \Delta w_{k,k',t}^i l_{k',t}^i \quad (22)$$

where each country-specific expectation $l_{k',t}^i$ is weighted by investor i 's share in total flows to k , $\sigma_{k,t}^i$, and by the *co-ownership linkage* within investor i , $\Delta w_{k,k',t}^i$, with

$$\Delta w_{k,k',t}^i = \left(\sum_{j=1}^{J(i)} \sigma_{k,t}^{i,j} \Delta w_{k',t}^{i,j} \right) \quad (23)$$

This linkage aggregates country k' 's relative allocations across investor i 's funds, weighting each fund by its importance in i 's flows to k . It captures the exposure of country k to country k' : investor i 's expectations about k' matter for k when the i -managed funds that invest in k also hold a disproportionately large share in k' .

Country granularity versus investor granularity Equation (22) expresses the granular term as an average of *investor-specific* expectations $l_{k',t}^i$ weighted by co-ownership linkages at the *investor* level. We now decompose it to separate the role of the global architecture of portfolios from that of specific investors.

To do so, notice that the idiosyncratic expectations $l_{k,t}^i$ are both country-specific and investor-specific. We therefore decompose them into the average country-specific component

of expectations across investors for country k , $l_{k,t} = (\sum_{i=1}^M l_{k,t}^i)/M$ and their investor-specific component $l_{k,t}^i - l_{k,t}$. We can then decompose the granular term into a term that is driven by “country granularity”, and two terms that are driven by “investor granularity”:

$$\begin{aligned} \Gamma_{k,t} = & \underbrace{\sum_{k'=1}^K \Delta w_{k,k',t} l_{k',t}}_{\Gamma_{k,t}^{country}} \\ & + \underbrace{\sum_{k'=1}^K \sum_{i=1}^M \sigma_{k,t}^i (\Delta w_{k,k',t}^i - \Delta w_{k,k',t}) (l_{k',t}^i - l_{k',t}) + \sum_{k'=1}^K \Delta w_{k,k',t} \sum_{i=1}^M \left(\sigma_{k,t}^i - \frac{1}{M} \right) (l_{k',t}^i - l_{k',t})}_{\Gamma_{k,t}^{investor}} \end{aligned} \quad (24)$$

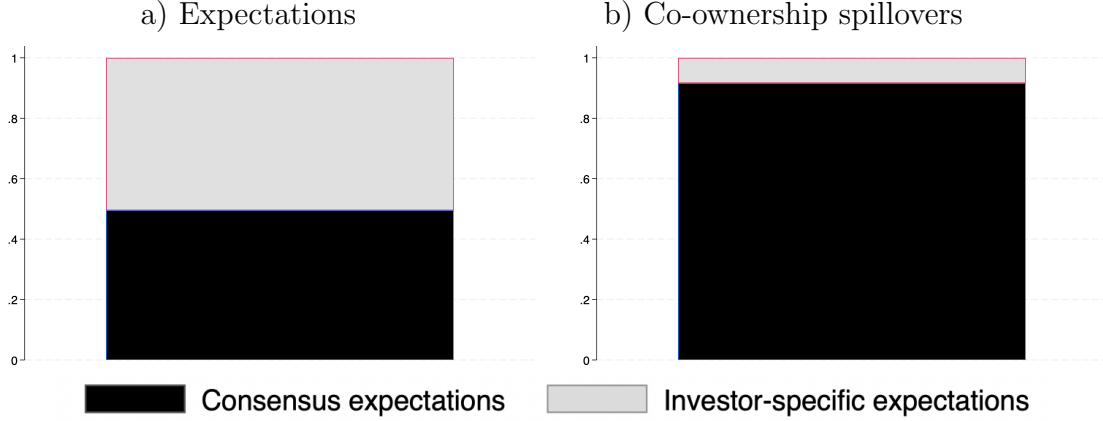
where $\Delta w_{k,k',t}$ is an average of the co-ownership linkages between country k and country k' across all investors, weighted by the importance of a given investor in the total flows to country k :

$$\Delta w_{k,k',t} = \left(\sum_{i=1}^M \sigma_{k,t}^i \Delta w_{k,k',t}^i \right) \quad (25)$$

The first term in Equation (24) shows that *consensus* expectations $l_{k',t}$ matter when the average co-ownership linkages are granular; we call it the “country granular component”, since its importance is driven by the granularity of countries in global portfolio shares. Investor-specific expectations $l_{k',t}^i - l_{k',t}$ enter through the remaining two terms: they matter when the linkages themselves are heterogeneous across investors, and when an investor’s contribution to country k ’s flows is granular. Together these form the “investor granular component”.

Figure 2 shows the relative contributions of consensus and investor-specific expectations, both to expectations $l_{k,t}^i$ themselves (Panel a)) and to co-ownership spillovers $\Gamma_{k,t}$ (Panel b)). Consensus and investor-specific expectations contribute almost equally to the variance of $l_{k,t}^i$. Yet consensus expectations account for more than 90% of the variance of $\Gamma_{k,t}$. This reflects the granularity of the associated weights, measured by their standard deviation (Gabaix, 2011; di Giovanni et al., 2014): the weights on the consensus expectations, the average co-ownership linkages $\Delta w_{k,k',t}$, are more dispersed by two orders of magnitude than those on the investor-specific expectations ($\sigma_{k,t}^i (\Delta w_{k,k',t}^i - \Delta w_{k,k',t})$ and $\Delta w_{k,k',t} (\sigma_{k,t}^i - 1/M)$), which explains the disproportionate contribution of the consensus (see Figure E.5 in the Appendix).

Figure 2: Variance decomposition of expectations and co-ownership spillovers



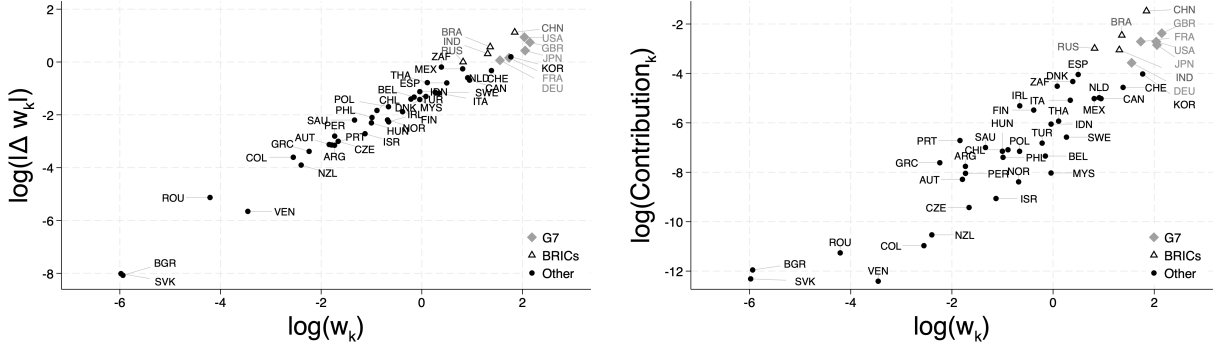
Notes: The figure represents the relative contributions of consensus expectations $l_{k,t}$ and investor-specific expectations $l_{k,t}^i - l_{k,t}$, to expectations $l_{k,t}^i$ themselves (Panel a)), and to co-ownership spillovers $\Gamma_{k,t}$ (Panel b)). In Panel a), the contribution of consensus expectations $l_{k,t}$ to expectations $l_{k,t}^i$ is measured as $V(l_{k,t})/V(l_{k,t}^i)$, and the contribution of investor-specific expectations $l_{k,t}^i - l_{k,t}$ to expectations $l_{k,t}^i$ is measured as $V(l_{k,t}^i - l_{k,t})/V(l_{k,t}^i)$. In Panel b), the contribution of consensus expectations to co-ownership spillovers $\Gamma_{k,t}$ is measured as $V(\Gamma_{k,t}^{country})/V(\Gamma_{k,t})$, and the contribution of investor-specific expectations to co-ownership spillovers $\Gamma_{k,t}$ is measured as $V(\Gamma_{k,t}^{investor})/V(\Gamma_{k,t})$.

Implications for external validity and data sparsity. This implies that co-ownership spillovers boil down, quantitatively, to the contribution of country granularity, so that only the global architecture of portfolios matters:

$$\Gamma_{k,t} \simeq \underbrace{\sum_{k'=1}^K \Delta w_{k,k',t} l_{k',t}}_{\Gamma_{k,t}^{country}} \quad (26)$$

This is reassuring for the external validity of our exercise: the aggregate co-ownership spillover is governed almost entirely by (i) consensus country shocks $l_{k'}$ and (ii) average co-ownership linkages $\Delta w_{k,k'}$. The contribution of investor-level heterogeneity in expectations and in portfolios is negligible. Quantifying the aggregate effect therefore does not require a full account of the universe of investors. It only requires that the observed portfolios deliver a representative estimate of $\Delta w_{k,k'}$ and that we observe consensus forecasts. Consistently, the variance-decomposition results based on consensus expectations instead of (imputed) investor-specific expectations (reported in Table D.5 in the Appendix) are very close to our baseline results.

Figure 3: Role of country average allocation



Notes: The left panel represents the scatter plot of the average allocations $w_k = \sum_{i=1}^M \sigma_k^i w_k^i$ against the log of the average absolute value of co-ownership linkages $|\Delta w_k| = \sum_{k'=1}^K \sigma_{k'} |\Delta w_{k',k}|$. The right panel represents the scatter plot of the average allocations $w_k = \sum_{i=1}^M \sigma_k^i w_k^i$ against the log of average contributions $\text{Contribution}_k = \sum_{k'=1}^K \sigma_{k'} \text{Contribution}_{k',k}$.

Contributors The variance of country k 's co-ownership spillovers is $\eta^2 V(\Gamma_k)$, with $V(\Gamma_k) = \sum_{l=1}^K (\Delta w_{k,l})^2 V(l_l)$. We therefore measure the contribution of country k' to country k 's spillovers as

$$\text{Contribution}_{k,k'} = \frac{(\Delta w_{k,k'})^2 V(l_{k'})}{\sum_{l=1}^K (\Delta w_{k,l})^2 V(l_l)} \quad (27)$$

We find that the large contributors are mostly countries with large allocations in portfolios. Figure E.6 in the appendix shows the average contribution of country k' across our sample, computed as $\text{Contribution}_{k'} = \sum_{k=1}^K \sigma_k \text{Contribution}_{k,k'}$. Among emerging economies, these are the BRICs (Brazil, Russian Federation, India, China), together with South Africa, South Korea and Mexico; among advanced economies, these are the main G7 countries: the UK, the US, France, Japan and Germany.

What drives these contributions is country size in portfolios, not expectation volatility. Figure E.7 in the appendix shows that the variation in contributions comes mostly from the average absolute co-ownership linkages $|\Delta w_{k'}| = \sum_{k=1}^K \sigma_k |\Delta w_{k,k'}|$, not from the idiosyncratic volatility $V(l_{k'})$: volatile expectations are not per se a systematic source of spillovers. Romania, Argentina, Greece and Venezuela have volatile expectations but contribute little. Because these average absolute linkages $|\Delta w_{k'}|$ are themselves highly correlated with the average portfolio shares $w_{k'}$, portfolio shares and contributions line up closely, as shown in Figure 3. Large countries are more likely to hold disproportionate shares in individual portfolios (the ten largest destinations alone account for 58% of holdings in our sample), which is the source of their large contribution.

Table 3: Capital flows, expectations, and the global financial cycle

	(1)	(2)
	$\log(A_{k,t}^{IMF})$	$\log(A_{k,t}^{EPFR})$
VARIABLES		
$\log VIX_t$	-0.054*** (0.013)	-0.113*** (0.032)
i_t^{US}	-0.036** (0.014)	-0.091*** (0.031)
$\beta \tilde{l}_{k,t}$	0.017*** (0.004)	0.089*** (0.010)
$\eta \Gamma_{k,t}$	0.035*** (0.006)	0.058*** (0.016)
Observations	2,440	2,974
Country FE	Yes	Yes
Within R-squared	0.11	0.12

Notes: The table reports quarterly regressions of (log) aggregate equity holdings on the co-ownership spillover $\eta \Gamma_{k,t}$, the excess-return reallocation term $\beta \tilde{l}_{k,t}$, the (log) VIX, and the US interest-rate stance i_t^{US} . The dependent variable is IMF IIP equity liabilities, $\log(A_{k,t}^{IMF})$ (column 1), or aggregate EPFR fund allocations to country k , $\log(A_{k,t}^{EPFR})$ (column 2). All variables are standardized, so coefficients are comparable across regressors. All specifications include country fixed effects. The reported fit is the within- R^2 . Standard errors clustered by quarter in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.3 Magnitude.

We have established that co-ownership spillovers make up a substantial part of *expectation-driven* capital-flow reallocation. How large is that reallocation relative to the forces the contagion literature has emphasized? Table 3 provides a benchmark. We regress aggregate country-level capital flows on the two expectation channels, $\eta \Gamma_{k,t}$ and $\beta \tilde{l}_{k,t}$, together with the two canonical drivers of the global financial cycle: the (log) VIX and the US interest-rate stance i_t^{US} . As two alternative measures of aggregate foreign equity holdings, we use EPFR equity holdings aggregated to the country-quarter level and the IMF IIP quarterly measure of equity liabilities. Because the VIX and the US interest rate are aggregate time series, the specification includes country fixed effects but no time fixed effects, and standard errors are clustered by quarter. All variables are standardized, so the coefficients measure the response of country-level flows, in standard deviations, to a one-standard-deviation shock, and are directly comparable across variables.

The co-ownership channel is of the same order of magnitude as the global drivers. Using the IMF holdings measure (column 1), a one-standard-deviation co-ownership shock moves country-level flows by 0.035 standard deviations, against 0.054 for the VIX and 0.036 for the US rate stance: about two-thirds of the former and on par with the latter (and twice the excess-return channel at 0.017). The same pattern holds with EPFR fund allocations (column 2), where the co-ownership channel (0.058) is about half the VIX (0.113) and two-thirds the US rate stance (0.091), while the excess-return channel (0.089) is higher. All coefficients are significant at the 1% level.

5 Conclusion

This paper has identified and quantified a channel of international financial contagion that operates through the granularity of country weights in delegated mutual fund portfolios. Linking the in-house GDP growth forecasts produced by global financial institutions to the assets under management and cross-country allocations of their equity mutual funds, we document a sharp asymmetry between the strong elasticity of fund flows and the weak elasticity of within-fund country weights, a reflection of portfolio stickiness. A delegated-portfolio model with sticky fund weights formalizes this asymmetry and decomposes expectation-driven capital flows into three channels: an excess-return channel, a hedging adjustment, and a *co-ownership spillover* channel. Plugging our estimates back into the model, co-ownership spillovers account for 60% of the variance of expectation-driven capital-flow reallocation, with the country-specific excess-return channel accounting for 40% and the hedging adjustment for less than 1%.

These findings carry several implications. First, because co-ownership spillovers are unrelated to recipient countries' fundamentals, they constitute a distinct source of cross-border capital misallocation that operates alongside the funding-shock channels traditionally emphasized in the contagion literature. An extensive literature documents the disruptions such capital misallocation can lead to.²² Second, the spatial pattern of transmission is Large-to-Small rather than North-to-South: the G7 and BRICS are the primary sources of spillovers, while small advanced and emerging economies are the primary recipients. Third, the quantitative result rests on the granularity of country weights, not on the granularity of investor wealth, and external validity requires only that the observed portfolios are representative of

²²Aggregate equity demand is inelastic, so fund-driven flows move prices (Gabaix and Koijen, 2021): for instance, Jotikasthira et al. (2012) find effects on emerging-market equity prices, country betas, and return comovement, and Camanho et al. (2022) on exchange rates. And capital inflows have first-order effects on output and credit (Blanchard et al., 2017), and they reshape the sectoral and firm-level allocation of resources, with lasting consequences for productivity (Benigno et al., 2015; Gopinath et al., 2017).

the global portfolio architecture. Policy-makers in small economies should therefore monitor not only the financial centers from which most capital originates, but also the large countries with which they share fund ownership.

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A Proofs

This appendix builds up to Proposition 2.2 in three steps. Appendix A.2 solves the investor's and the fund's optimization problems and derives the first-order conditions that underlie the country-level flow decomposition (Lemmas A.1 and A.2). Appendix A.3 combines these conditions with the Calvo-style updating friction to prove Proposition 2.1 and Corollary 2.1. Appendix A.4 aggregates the investor-fund flows to the country level (under the orthogonality and symmetry assumptions stated there) to prove Proposition 2.2, and Appendix A.5 proves the comparative statics of Corollary 2.2.

A.1 Some key constants

We define here some key constants for the analysis.

Definition A.1 (Key constants) *Define*

$$\begin{aligned}
V^i &= V(\mathcal{R}_p^i) \\
V^{i,j} &= Cov(R_p^{i,j}, R_p^{i,j} - \mathcal{R}_p^{i,j-}) \\
Cov^{i,j} &= Cov(R_p^{i,j}, \mathcal{R}_p^{i,j-}) / V^{i,j} \\
V_k^{i,j} &= Cov(R_k, R_k - R_{p,k-}^{i,j}) \\
Cov_k^{i,j} &= Cov(R_k, \mathcal{R}_p^{i,j-}) / V_k^{i,j} \\
\Delta Cov_k^{i,j} &= (Cov(R_k, R_{p,k-}^{i,j}) - Cov(R_k, \mathcal{R}_p^{i,j-})) / V_k^{i,j} \\
\widetilde{Cov}_k^{i,j} &= Cov_k^{i,j} - \Delta Cov_k^{i,j} Cov^{i,j} \\
\bar{E}^i(a_k^{i,j}) &= (\bar{E}^i(R_k) - r) / \gamma V_k^{i,j} - Cov_k^{i,j} (\bar{E}^i(\mathcal{R}_p^i) - r) / \gamma V^i - \Delta Cov_k^{i,j} \bar{E}^i(a^{i,j}) \\
\bar{E}^i(a^{i,j}) &= (\bar{E}^i(R_p^{i,j}) - r) / \gamma V^{i,j} - Cov^{i,j} (\bar{E}^i(\mathcal{R}_p^i) - r) / \gamma V^i \\
\bar{a}_k^{i,j} &= (\bar{E}^i(R_k) - r) / \gamma V_k^{i,j} - Cov_k^{i,j} (\bar{E}^i(\mathcal{R}_p^i) - r) / \gamma V^i
\end{aligned}$$

where $\mathcal{R}_p^i$ and $R_p^{i,j}$ are defined respectively by (1) and (2), $\mathcal{R}_p^{i,j-} = \sum_{j', j' \neq j} a^{i,j'} R_p^{i,j'} / \left(\sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} a^{i,j'} \right)$ is the return of the full portfolio of funds when excluding fund j , and $R_{p,k-}^{i,j} = \sum_{k', k' \neq k} w_k^{i,j} R_{k'}^{i,j} / (1 - w_k^{i,j})$ is the return of the fund portfolio when excluding country k .

A.2 Individual optimization

Investors and funds solve the same mean-variance program, at two nested stages. Investor i chooses fund allocations $a^i = (a^{i,1}, \dots, a^{i,\mathcal{J}(i)})'$ to maximize

$$U^i = E^i(\tilde{R}^i) - \frac{\gamma}{2} V(\tilde{R}^i), \quad \tilde{R}^i = \mathcal{R}_p^i \sum_{j=1}^{\mathcal{J}(i)} a^{i,j} + r \left(1 - \sum_{j=1}^{\mathcal{J}(i)} a^{i,j} \right), \quad (28)$$

where $\mathcal{R}_p^i$ is the return on the investor's portfolio of funds, defined in (1). Fund (i, j) , when it reoptimizes, chooses its country weights $w^{i,j}$ to maximize the same objective subject to $\sum_{k \in \mathcal{S}(i,j)} w_k^{i,j} = 1$, taking the distribution of country returns as given. We characterize the two allocations in turn.

The investor's fund allocation.

Lemma A.1 *Total equity investments must satisfy*

$$\sum_{j=1}^{\mathcal{J}(i)} a^{i,j} = \frac{E^i(\mathcal{R}_p^i) - r}{\gamma V^i} \quad (29)$$

with V^i is a constant term defined in A.1, and the optimal allocation to equity funds j $a^{i,j}$ must satisfy, for all $j = 1, \dots, \mathcal{J}(i)$,

$$a^{i,j} = \frac{E^i(R_p^{i,j}) - r}{\gamma V^{i,j}} - Cov^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} a^{i,j} \right) \quad (30)$$

where $V^{i,j}$ and $Cov^{i,j}$ are constant terms defined in A.1.

Proof. Maximizing (28) with respect to $a^{i,j}$, subject to (1), yields the following first-order

condition:

$$\begin{aligned}
E^i(R_p^{i,j}) - r &= \gamma \sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} Cov(R_p^{i,j}, R_p^{i,j'}) \\
&= \gamma \left(a^{i,j} V(R_p^{i,j}) + \left(\sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} - a^{i,j} \right) \sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} \frac{a^{i,j'}}{\sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} a^{i,j'}} Cov(R_p^{i,j}, R_p^{i,j'}) \right) \\
&= \gamma \left(a^{i,j} V(R_p^{i,j}) + \left(\sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} - a^{i,j} \right) Cov \left(R_p^{i,j}, \underbrace{\sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} \frac{a^{i,j'}}{\sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} a^{i,j'}} R_p^{i,j'}}_{\mathcal{R}_p^{i,j-}} \right) \right)
\end{aligned} \tag{31}$$

This yields Equation (30).

Equation (29) is obtained either by taking the sum of the above first-order condition across funds, weighted by $a^{i,j} / \sum_{j=1}^{\mathcal{J}(i)} a^{i,j}$, or by taking the derivative of (28) with respect to $\sum_{j=1}^{\mathcal{J}(i)} a^{i,j}$. ■

Equation (30) is the standard mean-variance allocation rule applied at the fund-allocation level: the investor loads on funds with high expected excess return, scaled by risk tolerance and variance, with a hedging adjustment for the covariance between fund j and the rest of the investor's portfolio (Markowitz, 1952; Lintner, 1965). Equation (29) pins down total equity demand from the same mean-variance logic applied to the aggregate portfolio. The variance and covariance terms $V^{i,j}$ and $Cov^{i,j}$ are treated as constants in the linearization; the symmetry condition (Assumption A.2, stated in Appendix A.4) further renders them homogeneous across countries within a fund.

The fund's country allocation.

Lemma A.2 *The optimal allocation $w_k^{i,j*}$ is such that $a_k^{i,j*} = w_k^{i,j*} a^{i,j}$, the total investment share of investor i that is channeled to country k through fund j , satisfies, for all $k \in \mathcal{S}(i, j)$*

$$a_k^{i,j*} = \frac{E^i(R_k) - r}{\gamma V_k^{i,j}} - Cov_k^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} a^{i,j} \right) - \Delta Cov_k^{i,j} a^{i,j} \tag{32}$$

where $V_k^{i,j}$, $Cov_k^{i,j}$ and $\Delta Cov_k^{i,j}$ are constant terms defined in A.1.

Proof. Maximizing terminal wealth $[\mathcal{R}_p^i (\sum_j a^{i,j}) + r (1 - \sum_j a^{i,j})] \Omega^i$ with respect to $w_k^{i,j}$, subject to (1) and (2), yields the following first-order condition for any $(k, K) \in \mathcal{S}(i, j)^2$ pair

of countries in the fund's portfolio:

$$\begin{aligned}
(E^i(R_k) - E^i(R_K)) &= \gamma \sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} \left(\sum_{k' \in \mathcal{S}(i,j')} w_{k'}^{i,j'} \text{Cov}(R_k, R_{k'}) - \sum_{k' \in \mathcal{S}(i,j)} w_{k'}^{i,j'} \text{Cov}(R_K, R_{k'}) \right) \\
&= \gamma \sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} \left(\sum_{k' \in \mathcal{S}(i,j')} w_{k'}^{i,j'} [\text{Cov}(R_k, R_{k'}) - \text{Cov}(R_K, R_{k'})] \right)
\end{aligned}$$

Noting that this is true for all $K \in \mathcal{S}(i, j)$, this can be written in vector form as follows:

$$Id(i, j) [E^i(R_k) - E^i(R)] = \gamma Id(i, j) (V_k^R - V^R) W^i a^i \quad (33)$$

where $W^i = (w^{i,1}, \dots, w^{i,j}, \dots, w^{i,\mathcal{J}(i)})$ is a $K \times \mathcal{J}(i)$ matrix of portfolio weights, $Id(i, j)$ is a $K \times K$ diagonal matrix, where the k^{th} element of the diagonal is equal to one if $k \in \mathcal{S}(i, j)$, and zero otherwise. For $k' \notin \mathcal{S}(i, j)$, $w_{k'}^{i,j} = 0$. Therefore, $(w^{i,j})' Id(i, j) = (w^{i,j})'$.

Left-multiplying by $(w^{i,j})'$, we obtain

$$E^i(R_k) - E^i(R_p^{i,j}) = \gamma (w^{i,j})' (V_k^R - V^R) W^i a^i \quad (34)$$

Note that (31) can also be written in a vector form:

$$E^i(R_p^{i,j}) - r = \gamma (w^{i,j})' V^R W^i a^i$$

Substituting into (34), we obtain

$$\begin{aligned}
E^i(R_k) - r &= \gamma(w^{i,j})' V_k^R W^i a^i \\
&= \gamma \sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} \sum_{k' \in \mathcal{S}(i,j')} w_{k'}^{i,j'} \text{Cov}(R_k, R_{k'}) \\
&= \gamma \left(a^{i,j} \sum_{k' \in \mathcal{S}(i,j)} w_{k'}^{i,j} \text{Cov}(R_k, R_{k'}) + \underbrace{\left(\sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} - a^{i,j} \right) \sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} \frac{a^{i,j'}}{\sum_{j'=1, j' \neq j}^{\mathcal{J}(i)} a^{i,j'}} \sum_{k' \in \mathcal{S}(i,j')} w_{k'}^{i,j'} \text{Cov}(R_k, R_{k'})}_{\text{Cov}(R_k, \mathcal{R}_p^{i,j-})} \right) \\
&= \gamma \left(a^{i,j} \left(w_k^{i,j} V(R_k) + (1 - w_k^{i,j}) \underbrace{\sum_{k' \in \mathcal{S}(i,j), k' \neq k} \frac{w_{k'}^{i,j}}{1 - w_k^{i,j}} \text{Cov}(R_k, R_{k'})}_{\text{Cov}(R_k, \mathcal{R}_{p,k-}^{i,j})} \right) + \left(\sum_{j'=1}^{\mathcal{J}(i)} a^{i,j'} - a^{i,j} \right) \text{Cov}(R_k, \mathcal{R}_p^{i,j-}) \right)
\end{aligned}$$

This yields Equation (32). ■

The first two terms of (32) parallel (30): the fund loads on country k in proportion to its expected excess return scaled by risk tolerance and variance, with a hedging adjustment for the covariance between k and the rest of the investor's portfolio. The third term is the new piece introduced by the delegated structure. It is proportional to $a^{i,j}$ and loads on $-\Delta \text{Cov}_k^{i,j}$, which can be read as the fixed share of investor i 's allocation to fund j that is routed to country k to hedge exposure to fund j itself. As $a^{i,j}$ rises, the investor's exposure to the fund rises, while the exposure to the rest of her portfolio falls; the fund therefore allocates less to k when k is a poor hedge for the fund and more when k is a poor hedge for the investor's wider portfolio.

A.3 Proof of Proposition 2.1 and Corollary 2.1

We follow similar steps as in the proof of Lemma A.2 to derive the default shares $\bar{w}_k^{i,j}$, taking into account the fact that the fund investments $a^{i,j}$ are not known:

$$\bar{w}^{i,j} = \frac{\bar{E}^i(R_k) - r}{\gamma \bar{V}_k^{i,j} \bar{E}^i(a^{i,j})} - \frac{\overline{\text{Cov}}_k^{i,j} \left(\sum_{j'=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j'}) \right)}{\bar{E}^i(a^{i,j})} - \frac{\Delta \overline{\text{Cov}}_k^{i,j}}{\bar{E}^i(a^{i,j})}$$

with $\bar{V}_k^{i,j} = \overline{Cov}(R_k, R_k - R_{p,k-}^{i,j})$, $\overline{Cov}_k^{i,j} = \overline{Cov}(R_k, \mathcal{R}_p^{i,j-})/\bar{V}_k^{i,j}$ and $\overline{\Delta Cov}_k^{i,j} = (\overline{Cov}(R_k, R_{p,k-}^{i,j}) - \overline{Cov}(R_k, \mathcal{R}_p^{i,j-}))/\bar{V}_k^{i,j}$. $\bar{V}(\cdot)$ and $\overline{Cov}(\cdot)$ are the variance and covariance conditional on the beginning-of-period information $\bar{\mathcal{I}}^i$. Under Assumption 2.1, these terms can be replaced by their end-of-period counterparts:

$$\bar{w}^{i,j} = \frac{\bar{E}^i(R_k) - r}{\gamma V_k^{i,j} \bar{E}^i(a^{i,j})} - Cov_k^{i,j} \frac{\left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j})\right)}{\bar{E}^i(a^{i,j})} - \Delta Cov_k^{i,j} \quad (35)$$

Using the definition of $a_k^{i,j}$, (3), the optimal updated and ex ante allocations, (32), and Equation (35), we obtain:

$$\begin{aligned} a_k^{i,j} &= p \left(\frac{E^i(R_k) - r}{\gamma V_k^{i,j}} - Cov_k^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} a^{i,j} \right) - \Delta Cov_k^{i,j} a^{i,j} \right) \\ &\quad + (1-p) \left(\frac{\bar{E}^i(R_k) - r}{\gamma V_k^{i,j} \bar{E}^i(a^{i,j})} - Cov_k^{i,j} \frac{\left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j})\right)}{\bar{E}^i(a^{i,j})} - \Delta Cov_k^{i,j} \right) a^{i,j} \\ &= p \left(\frac{E^i(R_k) - r}{\gamma V_k^{i,j}} - Cov_k^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} a^{i,j} \right) \right) \\ &\quad + (1-p) \left(\frac{\bar{E}^i(R_k) - r}{\gamma V_k^{i,j} \bar{E}^i(a^{i,j})} - Cov_k^{i,j} \frac{\left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j})\right)}{\bar{E}^i(a^{i,j})} \right) a^{i,j} - \Delta Cov_k^{i,j} a^{i,j} \end{aligned} \quad (36)$$

We take the beginning-of-period expectation, and subtract it:

$$\begin{aligned} a_k^{i,j} - \bar{E}^i(a_k^{i,j}) &= p \left(\frac{E^i(r_k)}{\gamma V_k^{i,j}} - Cov_k^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} a^{i,j} - \sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j}) \right) \right) \\ &\quad + (1-p) \left(\frac{\bar{E}^i(R_k) - r}{\gamma V_k^{i,j} \bar{E}^i(a^{i,j})} - Cov_k^{i,j} \frac{\left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j})\right)}{\bar{E}^i(a^{i,j})} \right) (a^{i,j} - \bar{E}^i(a^{i,j})) \\ &\quad - \Delta Cov_k^{i,j} (a^{i,j} - \bar{E}^i(a^{i,j})) \end{aligned}$$

Using (29) and (30), we obtain:

$$\begin{aligned}
a_k^{i,j} - \bar{E}^i(a_k^{i,j}) &= p \left(\frac{E^i(r_k)}{\gamma V_k^{i,j}} - Cov_k^{i,j} \frac{E^i(\mathbf{r}_p^i)}{\gamma V^i} \right) \\
&\quad + (1-p) \left(\frac{\bar{E}^i(R_k) - r}{\gamma V_k^{i,j} \bar{E}^i(a^{i,j})} - Cov_k^{i,j} \frac{\left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j}) \right)}{\bar{E}^i(a^{i,j})} \right) \left(\frac{E^i(r_p^{i,j})}{\gamma V^{i,j}} - Cov^{i,j} \frac{E^i(\mathbf{r}_p^i)}{\gamma V^i} \right) \\
&\quad - \Delta Cov_k^{i,j} \left(\frac{E^i(r_p^{i,j})}{\gamma V^{i,j}} - Cov^{i,j} \frac{E^i(\mathbf{r}_p^i)}{\gamma V^i} \right) \\
&= p \frac{E^i(r_k)}{\gamma V_k^{i,j}} + (1-p) \frac{\bar{a}_k^{i,j}}{\bar{E}^i(a^{i,j})} \frac{E^i(r_p^{i,j})}{\gamma V^{i,j}} - \Delta Cov_k^{i,j} \frac{E^i(r_p^{i,j})}{\gamma V^{i,j}} \\
&\quad - (Cov_k^{i,j} - \Delta Cov_k^{i,j} Cov^{i,j}) \frac{E^i(\mathbf{r}_p^i)}{\gamma V^i} - (1-p) \left(\frac{\bar{a}_k^{i,j}}{\bar{E}^i(a^{i,j})} Cov^{i,j} - Cov_k^{i,j} \right) \frac{E^i(\mathbf{r}_p^i)}{\gamma V^i}
\end{aligned}$$

with $\bar{a}_k^{i,j} = (\bar{E}^i(R_k) - r)/\gamma V_k^{i,j} - Cov_k^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j}) \right)$.

Finally, the beginning-of-period expectation of $a_k^{i,j}$ obtained from (36) is $\bar{E}^i(a_k^{i,j}) = (\bar{E}^i(R_k) - r)/\gamma V_k^{i,j} - Cov_k^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j}) \right) - \Delta Cov_k^{i,j} \bar{E}^i(a^{i,j})$. Then we take the beginning-of-period expectations of $a^{i,j}$ and $\sum_{j=1}^{\mathcal{J}(i)} a^{i,j}$ by using (30) and (29) and obtain $\bar{E}^i(a^{i,j}) = (\bar{E}^i(R_p^{i,j}) - r)/\gamma V^{i,j} - Cov^{i,j} \left(\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j}) \right)$ and $\sum_{j=1}^{\mathcal{J}(i)} \bar{E}^i(a^{i,j}) = (\bar{E}^i(\mathcal{R}_p^i) - r)/\gamma V^i$. This yields Proposition 2.1. Collecting the terms that multiply $E^i(r_k)$, $E^i(r_p^{i,j})$, and $E^i(\mathbf{r}_p^i)$ and normalizing by $\bar{E}^i(a_k^{i,j})$ yields the elasticities $\beta_k^{i,j}$, $\delta_k^{i,j}$, and $\theta_k^{i,j}$ of Corollary 2.1; their explicit expressions, together with the decomposition $\delta_k^{i,j} = \eta_k^{i,j} - \phi_k^{i,j} \Delta Cov_k^{i,j}$, are derived in (43) within the proof of Lemma A.4.

A.4 Proof of Proposition 2.2

The proof relies on two technical assumptions, which we state here.

Assumption A.1 (Orthogonality) For all $i = 1, \dots, M$ and for all $k = 1, \dots, K$, $\sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (\theta_k^{i,j} - \theta_k^i) \Gamma^{i,j}$, $\sum_{i=1}^M \sigma_k^i (\beta_k^i - \beta_k) l_k^i$, $\sum_{i=1}^M \sigma_k^i (\delta_k^i - \delta_k) \left(\sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right)$, $\sum_{i=1}^M \sigma_k^i (\theta_k^i - \theta_k) \Gamma^i$, $\sum_{i=1}^M \sigma_k^i (\beta_k^i + \delta_k^i + \theta_k^i - \beta_k - \delta_k - \theta_k) W^i$, $\sum_{k=1}^K \sigma_k (\beta_k - \beta) l_k$ and $\sum_{k=1}^K \sigma_k (\delta_k - \delta) \Gamma_k$ are small relative to $\sum_{k \in \mathcal{S}(i,j)} w_k^{i,j} l_k^i$.

Assumption A.2 (Symmetry) For all $i = 1, \dots, M$, $j = 1, \dots, \mathcal{J}(i)$ and $(k, k') \in \mathcal{S}(i, j)^2$, $k \neq k'$:

$$(a) \quad \bar{E}^i(R_k) \simeq \bar{E}^i(R_{k'});$$

- (b) $Cov(R_k, \mathcal{R}_p^{i,j-}) \simeq Cov(R_{k'}, \mathcal{R}_p^{i,j-})$ and $Cov(R_k, R_{p,k-}^{i,j}) \simeq Cov(R_{k'}, R_{p,k'-}^{i,j})$;
- (c) $\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \left(\frac{\bar{E}^i(A_k^{i,j})}{\bar{E}(A_k)} - \frac{\bar{E}^i(A^{i,j})}{\bar{E}(A)} \right) x^{i,j} \simeq 0$, where $x^{i,j} = \sum_{k \in \mathcal{S}(i,j)} w_k^{i,j} x_k^{i,j}$ for all $x_k^{i,j} \in \{\beta_k^{i,j}, \eta_k^{i,j}, (\phi \Delta Cov)_k^{i,j}, \delta_k^{i,j}, \theta_k^{i,j}\}$.

Assumption A.1 ensures that the granular residual remains relevant while allowing useful approximations during aggregation: the elasticity coefficients are required to be orthogonal to portfolio shares and expectations. Assumption A.2 ensures that the elasticity coefficients can be treated as homogeneous across countries: two countries in a given fund must have (a) sufficiently similar ex-ante return expectations and (b) sufficiently similar hedging properties, and (c) fund-level elasticities must be unrelated to a fund's contribution to total country flows.

We first prove the following lemma:

Lemma A.3 (Aggregation) *We assume that Assumptions 2.2 and A.1 are satisfied. In that case, Equation (8) can be written as:*

$$\begin{aligned} \frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} = & \underbrace{\beta_k \left(\sum_{i=1}^M \sigma_k^i (l_k^i - \Gamma^i) \right) + \delta_k \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (\Gamma^{i,j} - \Gamma^i) \right) + (\Theta_k - \Theta) \left(\sum_{i=1}^M \sigma^i (W^i + \Gamma^i) \right)}_{\tilde{\Delta} a_k} \\ & + \underbrace{\Theta \left(\sum_{i=1}^M \sigma^i (W^i + \Gamma^i) \right)}_{\Delta a} \end{aligned} \quad (37)$$

where $\Theta_k = \beta_k + \delta_k + \theta_k$ is the sum of the country-, fund- and investor-level elasticities, $\sigma^i = \bar{E}^i(a^i) \Omega^i / \bar{E}(a) \Omega = \bar{E}^i(A^i) / \bar{E}(A)$ is the ex-ante share of investor i in total equity investments.

Proof. Using Assumption 2.2, the surprise capital flows (5) admit the following decomposition:

$$\frac{a_k^{i,j} - \bar{E}^i(a_k^{i,j})}{\bar{E}^i(a_k^{i,j})} = \beta_k^{i,j} l_k^i + \delta_k^{i,j} \Gamma^{i,j} + \theta_k^{i,j} \Gamma^i + \Theta_k^{i,j} W^i$$

with $\Theta_k^{i,j} = \beta_k^{i,j} + \delta_k^{i,j} + \theta_k^{i,j}$.

We replace in Equation (8):

$$\begin{aligned} \frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} &= \sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (\beta_k^{i,j} l_k^i + \delta_k^{i,j} \Gamma^{i,j} + \theta_k^{i,j} \Gamma^i + \Theta_k^{i,j} W^i) \\ &= \sum_{i=1}^M \sigma_k^i \beta_k^i l_k^i + \sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \delta_k^{i,j} \Gamma^{i,j} + \sum_{i=1}^M \sigma_k^i \theta_k^i \Gamma^i + \sum_{i=1}^M \sigma_k^i \Theta_k^i W^i \end{aligned}$$

Note that

$$\sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \delta_k^{i,j} \Gamma^{i,j} = \delta_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} + \underbrace{\sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (\delta_k^{i,j} - \delta_k^i) \Gamma^{i,j}}_{\simeq 0}$$

where we used Assumption A.1. Therefore:

$$\frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} \simeq \sum_{i=1}^M \sigma_k^i \beta_k^i l_k^i + \sum_{i=1}^M \sigma_k^i \delta_k^i \left(\sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right) + \sum_{i=1}^M \sigma_k^i \theta_k^i \Gamma^i + \sum_{i=1}^M \sigma_k^i \Theta_k^i W^i$$

Take the first term:

$$\sum_{i=1}^M \sigma_k^i \beta_k^i l_k^i = \beta_k \sum_{i=1}^M \sigma_k^i l_k^i + \underbrace{\sum_{i=1}^M \sigma_k^i (\beta_k^i - \beta_k) l_k^i}_{\simeq 0}$$

where we used Assumption A.1 again. We apply similar steps to the other terms, and we obtain, using Assumption A.1:

$$\frac{a_k - \bar{E}(a_k)}{\bar{E}(a_k)} \simeq \beta_k \left(\sum_{i=1}^M \sigma_k^i l_k^i \right) + \delta_k \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right) + \theta_k \left(\sum_{i=1}^M \sigma_k^i \Gamma^i \right) + \Theta_k \left(\sum_{i=1}^M \sigma_k^i W^i \right) \quad (38)$$

We aggregate the country flows using Equation (38):

$$\begin{aligned}
& \frac{a - \bar{E}(a)}{\bar{E}(a)} \\
& \simeq \sum_{k=1}^K \sigma_k \left(\beta_k \left(\sum_{i=1}^M \sigma_k^i l_k^i \right) + \delta_k \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right) + \theta_k \left(\sum_{i=1}^M \sigma_k^i \Gamma^i \right) + \Theta_k \left(\sum_{i=1}^M \sigma_k^i W^i \right) \right) \\
& \simeq \sum_{k=1}^K \sigma_k \beta_k \left(\sum_{i=1}^M \sigma_k^i l_k^i \right) + \sum_{k=1}^K \sigma_k \delta_k \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right) + \sum_{k=1}^K \sigma_k \theta_k \left(\sum_{i=1}^M \sigma_k^i \Gamma^i \right) + \sum_{k=1}^K \sigma_k \Theta_k \left(\sum_{i=1}^M \sigma_k^i W^i \right)
\end{aligned}$$

Take the first term:

$$\begin{aligned}
\sum_{k=1}^K \sigma_k \beta_k \left(\sum_{i=1}^M \sigma_k^i l_k^i \right) &= \beta \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^M \sigma_k^i l_k^i \right) + \sum_{k=1}^K \sigma_k (\beta_k - \beta) \left(\sum_{i=1}^M \sigma_k^i l_k^i \right) \\
&= \beta \sum_{i=1}^M \sum_{k=1}^K \sigma_k \sigma_k^i l_k^i + \underbrace{\sum_{k=1}^K \sigma_k (\beta_k - \beta) l_k}_{\simeq 0} \\
&\simeq \beta \sum_{i=1}^M \sigma^i \sum_{k=1}^K w_k^i l_k^i = \beta \sum_{i=1}^M \sigma^i \Gamma^i
\end{aligned}$$

Take the second term:

$$\begin{aligned}
\sum_{k=1}^K \sigma_k \delta_k \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right) &= \delta \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right) + \sum_{k=1}^K \sigma_k (\delta_k - \delta) \left(\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} \Gamma^{i,j} \right) \\
&= \delta \sum_{k=1}^K \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \sigma_k \sigma_k^i \sigma_k^{i,j} \Gamma^{i,j} + \underbrace{\sum_{k=1}^K \sigma_k (\delta_k - \delta) \Gamma_k}_{\simeq 0} \\
&\simeq \delta \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \sigma^i \sigma^{i,j} \Gamma^{i,j} \underbrace{\sum_{k \in \mathcal{S}(i,j)} w_k^{i,j}}_{=1} \\
&\simeq \delta \sum_{i=1}^M \sigma^i \sum_{j=1}^{\mathcal{J}(i)} \sigma^{i,j} \sum_{k \in \mathcal{S}(i,j)} w_k^{i,j} l_k^i \\
&\simeq \delta \sum_{i=1}^M \sigma^i \sum_{j=1}^{\mathcal{J}(i)} \sum_{k \in \mathcal{S}(i,j)} \sigma^{i,j} w_k^{i,j} l_k^i \\
&\simeq \delta \sum_{i=1}^M \sigma^i \sum_{k=1}^K l_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma^{i,j} w_k^{i,j} \\
&\simeq \delta \sum_{i=1}^M \sigma^i \sum_{k=1}^K w_k^i l_k^i \\
&\simeq \delta \sum_{i=1}^M \sigma^i \Gamma^i
\end{aligned}$$

Take the third term:

$$\begin{aligned}
\sum_{k=1}^K \sigma_k \theta_k \left(\sum_{i=1}^M \sigma_k^i \Gamma^i \right) &= \theta \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^M \sigma_k^i \Gamma^i \right) + \sum_{k=1}^K \sigma_k (\theta_k - \theta) \left(\sum_{i=1}^M \sigma_k^i \Gamma^i \right) \\
&= \theta \sum_{k=1}^K \sum_{i=1}^M \sigma_k \sigma_k^i \Gamma^i + \underbrace{\sum_{k=1}^K \sigma_k (\theta_k - \theta) \Gamma_k}_{\simeq 0} \\
&\simeq \theta \sum_{i=1}^M \Gamma^i \sum_{k=1}^K \sigma_k \sigma_k^i \\
&\simeq \theta \sum_{i=1}^M \sigma^i \Gamma^i
\end{aligned}$$

We follow similar steps for the fourth term and find

$$\sum_{k=1}^K \sigma_k \Theta_k \left(\sum_{i=1}^M \sigma_k^i W^i \right) \simeq \Theta \sum_{i=1}^M \sigma^i W^i$$

Noting that $\Theta = \beta + \delta + \theta$, aggregate capital flows can be written as

$$\frac{a - \bar{E}(a)}{\bar{E}(a)} \simeq \Theta \sum_{i=1}^M \sigma^i (W^i + \Gamma^i) \quad (39)$$

Combining (38) and (39) yields the decomposition (37).²³ ■

We now prove the following lemma:

Lemma A.4 (Elasticity homogeneity) *Under Assumption A.2, $\beta_k \simeq \beta$, $\eta_k \simeq \eta$, $(\phi \Delta Cov)_k \simeq (\phi \Delta Cov)$, $\delta_k \simeq \delta$ and $\Theta_k \simeq \Theta$. Additionally, $\beta \propto p$ and $\delta = \eta - (\phi \Delta Cov)$ with $\eta \propto 1 - p$.*

Proof. Consider Assumption A.2. Denote

$$\begin{aligned} \Delta^i &\simeq \bar{E}^i(R_k) \simeq \bar{E}^i(R_p^{i,j}) \simeq \bar{E}^i(\mathcal{R}_p^i) \\ \rho^{i,j} &\simeq Cov(R_k, \mathcal{R}_p^{i,j-}) \simeq Cov(R_p^{i,j}, \mathcal{R}_p^{i,j-}) \end{aligned}$$

Then, notice that

$$V_k^{i,j} \bar{a}_k^{i,j} \simeq V^{i,j} \bar{E}^i(a^{i,j}) \simeq \frac{\Delta^i}{\gamma} \left(1 - \frac{\rho^{i,j}}{V^i} \right) \quad (40)$$

Similarly,

$$V_k^{i,j} Cov_k^{i,j} \simeq V^{i,j} Cov^{i,j} \simeq \rho^{i,j} \quad (41)$$

Note that Assumption A.2 implies $V_k^{i,j} \Delta Cov_k^{i,j} = V_{k'}^{i,j} \Delta Cov_{k'}^{i,j}$ for all $k' \neq k$. Let's denote

$$V_k^{i,j} \Delta Cov_k^{i,j} = \Delta \rho^{i,j} \quad (42)$$

²³Passing from the country-level growth term $\Theta \sum_i \sigma_k^i (W^i + \Gamma^i)$ to the investor-level term $\Theta \sum_i \sigma^i (W^i + \Gamma^i)$ in (37) drops the residual $\Theta \sum_{i=1}^M (\sigma_k^i - \sigma^i) (W^i + \Gamma^i)$, which vanishes under Assumption A.2(c).

Now, we can rewrite the coefficients as follows, using Corollary 2.1:

$$\begin{aligned}
\beta_k^{i,j} &= \frac{p}{\gamma V_k^{i,j} \bar{E}^i(a_k^{i,j})} \simeq \frac{p}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})} = \beta^{i,j} \\
\eta_k^{i,j} &= (1-p) \frac{\bar{a}_k^{i,j}}{\gamma V^{i,j} \bar{E}^i(a^{i,j}) \bar{E}^i(a_k^{i,j})} = (1-p) \frac{V_k^{i,j} \bar{a}_k^{i,j}}{\gamma V^{i,j} \bar{E}^i(a^{i,j}) V_k^{i,j} \bar{E}^i(a_k^{i,j})} \\
&\simeq \frac{1-p}{\gamma V_k^{i,j} \bar{E}^i(a_k^{i,j})} \simeq \frac{1-p}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})} = \eta^{i,j} \\
\phi_k^{i,j} \Delta Cov_k^{i,j} &= \frac{\Delta Cov_k^{i,j}}{\gamma V^{i,j} \bar{E}^i(a_k^{i,j})} = \frac{V_k^{i,j} \Delta Cov_k^{i,j}}{\gamma V^{i,j} V_k^{i,j} \bar{E}^i(a_k^{i,j})} = \frac{\Delta \rho^{i,j}}{V^{i,j} \left(\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j}) \right)} \\
&= (\phi \Delta Cov)^{i,j} \\
\delta_k^{i,j} &= \eta_k^{i,j} - \phi_k^{i,j} \Delta Cov_k^{i,j} \simeq \eta^{i,j} - (\phi \Delta Cov)^{i,j} = \delta^{i,j} \\
\theta_k^{i,j} &= - \frac{\widetilde{Cov}_k^{i,j}}{\gamma V^i \bar{E}^i(a_k^{i,j})} - (1-p) \frac{Cov^{i,j} \bar{a}_k^{i,j} / \bar{E}^i(a^{i,j}) - Cov_k^{i,j}}{\gamma V^i \bar{E}^i(a_k^{i,j})} \simeq - \frac{\widetilde{Cov}_k^{i,j}}{\gamma V^i \bar{E}^i(a_k^{i,j})} \\
&\simeq - \frac{Cov_k^{i,j} - Cov^{i,j} \Delta Cov_k^{i,j}}{\gamma V^i \bar{E}^i(a_k^{i,j})} \simeq - \frac{V_k^{i,j} Cov_k^{i,j} - Cov^{i,j} V_k^{i,j} \Delta Cov_k^{i,j}}{\gamma V^i V_k^{i,j} \bar{E}^i(a_k^{i,j})} \\
&\simeq - \frac{\rho^{i,j} - Cov^{i,j} \Delta \rho^{i,j}}{V^i \left(\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j}) \right)} = \theta^{i,j}
\end{aligned} \tag{43}$$

which also implies that $\Theta_k^{i,j} = \beta_k^{i,j} + \delta_k^{i,j} + \theta_k^{i,j} \simeq \beta^{i,j} + \delta^{i,j} + \theta^{i,j} \simeq \Theta^{i,j}$. Within a fund, all the coefficients are homogeneous across countries.

We now aggregate the country-specific coefficients across funds. For country $k = 1, \dots, K$, and for $x = \{\beta, \delta, \eta, \phi \Delta Cov, \Theta\}$, we have

$$x_k = \sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} x_k^{i,j} \simeq \sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} x^{i,j} \simeq x + \sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (x^{i,j} - x)$$

where

$$x = \sum_{k=1}^K \sigma_k x_k$$

Consider the second term:

$$\begin{aligned}
\sum_{i=1}^M \sigma_k^i \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} (x^{i,j} - x) &= \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^i \sigma_k^{i,j} x^{i,j} - \sum_{k'=1}^K \sigma_{k'} \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \sigma_{k'}^i \sigma_{k'}^{i,j} x^{i,j} \\
&= \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \underbrace{\sigma_k^i \sigma_k^{i,j}}_{\frac{\bar{E}^i(a_k^{i,j})\Omega^i}{\bar{E}(a_k)\Omega}} x^{i,j} - \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} x^{i,j} \sum_{k'=1}^K \underbrace{\sigma_{k'}^i \sigma_{k'}^{i,j}}_{\frac{\bar{E}^i(a_{k'}^{i,j})\Omega^i}{\bar{E}(a_{k'})\Omega}} \\
&\quad \underbrace{\frac{\bar{E}^i(a_{k'}^{i,j})\Omega^i}{\bar{E}(a_{k'})\Omega}}_{\frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega}} \\
&= \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \left(\frac{\bar{E}^i(a_k^{i,j})\Omega^i}{\bar{E}(a_k)\Omega} - \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \right) x^{i,j} \\
&= \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \left(\frac{\bar{E}^i(A_k^{i,j})}{\bar{E}(A_k)} - \frac{\bar{E}^i(A^{i,j})}{\bar{E}(A)} \right) x^{i,j} \\
&\simeq 0
\end{aligned}$$

where we used Assumption A.2. Therefore, $x_k \simeq x$, for $x = \{\beta, \delta, \eta, \phi\Delta Cov, \Theta\}$. This proves coefficient homogeneity.

Now, note that, for $x = \{\beta, \delta, \eta, \phi\Delta Cov, \Theta\}$:

$$\begin{aligned}
x &= \sum_{k=1}^K \sigma_k x_k = \sum_{k=1}^K \sigma_k \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \sigma_k^{i,j} x^{i,j} = \sum_{k=1}^K \sigma_k \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a_k^{i,j})\Omega^i}{\bar{E}(a_k)\Omega} x^{i,j} \\
&= \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} x^{i,j} + \underbrace{\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \left(\frac{\bar{E}^i(a_k^{i,j})\Omega^i}{\bar{E}(a_k)\Omega} - \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \right) x^{i,j}}_{\simeq 0} \right) \\
&= \sum_{k=1}^K \sigma_k \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} x^{i,j} = \underbrace{\left(\sum_{k=1}^K \sigma_k \right)}_{=1} \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} x^{i,j} = \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} x^{i,j}
\end{aligned} \tag{44}$$

Using the expression for $\beta^{i,j}$ and $\eta^{i,j}$ in (43), we obtain:

$$\begin{aligned}\beta &= \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{p}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})} \\ &= p \left(\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{1}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})} \right) \propto p\end{aligned}\quad (45)$$

$$\begin{aligned}\eta &= \sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{1-p}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})} \\ &= (1-p) \left(\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{1}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})} \right) \propto 1-p\end{aligned}\quad (46)$$

Moreover, since $\delta_k^{i,j} = \eta_k^{i,j} - \phi_k^{i,j} \Delta Cov_k^{i,j} = \eta_k^{i,j} - (\phi \Delta Cov)_k^{i,j}$, then $\delta = \eta - (\phi \Delta Cov)$.

■

Combining Lemma A.3 and A.4, we obtain decomposition (12) of Proposition 2.2.

A.5 Proof of Corollary 2.2

Points (i) and (ii) derive directly from Proposition 2.2, which states that $\beta \propto p$ and $\delta = \eta - (\phi \Delta Cov)$ with $\eta \propto 1-p$. Since $\beta \propto p$, β is decreasing in $1-p$, which is point (i). Since $\eta \propto 1-p$ and $(\phi \Delta Cov)$ is independent of p (as shown for point (iii) below), $\delta = \eta - (\phi \Delta Cov)$ is increasing in $1-p$ and positive when $1-p$ is large enough, which is point (ii).

Point (iii) can be derived as follows. Note that $\Theta = \beta + \delta + \theta = \beta + \eta - (\phi \Delta Cov) + \theta$. Consider $\beta + \eta$, $(\phi \Delta Cov)$ and θ separately. First, using (45) and (46), we obtain:

$$\beta + \eta = \underbrace{(p + 1 - p)}_{=1} \underbrace{\left(\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{1}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i}\right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})} \right)}_{\text{Independent from } p}$$

Second, using expression (44) with $x = \{\phi \Delta Cov, \theta\}$, along with the expressions for $(\phi \Delta Cov)^{i,j}$

and $\theta^{i,j}$ in (43)), we obtain:

$$\begin{aligned}
(\phi \Delta Cov) &= \underbrace{\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{\Delta \rho^{i,j}}{V^{i,j} \left(\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i} \right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j}) \right)}}_{\text{Independent from } p} \\
\theta &= \underbrace{\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} - \frac{\rho^{i,j} - Cov^{i,j} \Delta \rho^{i,j}}{V^i \left(\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i} \right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j}) \right)}}_{\text{Independent from } p}
\end{aligned}$$

Therefore, Θ is independent from p .

To show point (iv), we take the ratio of β to η using (45) and (46):

$$\frac{\beta}{\eta} = \frac{p}{1-p} \underbrace{\frac{\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{1}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i} \right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})}}{\sum_{i=1}^M \sum_{j=1}^{\mathcal{J}(i)} \frac{\bar{E}^i(a^{i,j})\Omega^i}{\bar{E}(a)\Omega} \frac{1}{\Delta^i \left(1 - \frac{\rho^{i,j}}{V^i} \right) - \gamma \Delta \rho^{i,j} \bar{E}^i(a^{i,j})}}}_{=1}$$

B Data Appendix

B.1 Estimation of $\Delta Cov_k^{i,j}$

According to Lemma A.2, $\Delta Cov_k^{i,j}$ is the difference between the scaled covariance of the country return k with the fund-level return excluding country k $Cov(R_k, R_{p,k-}^{i,j})/V_k^{i,j}$ and the scaled covariance of the country return k with the investor-level return excluding fund j and $Cov(R_k, \mathcal{R}_{p,j-}^{i,j})/V_k^{i,j}$, where $V_k^{i,j} = Cov(R_k, R_k - R_{p,k-}^{i,j})$. We proxy for these scaled covariances by using the country equity MSCI return data.

Define the aggregate fund-level return, the aggregate fund-level return excluding country

k and the aggregate investor-level return excluding fund j respectively as follows:

$$\begin{aligned}
R_{p,k-,t}^{i,j} &= \sum_{l \neq k, l \in \mathcal{S}(i,j)} \frac{w_{l,t}^{i,j}}{\sum_{l \neq k, l \in \mathcal{S}(i,j)} w_{l,t}^{i,j}} R_{l,t}, \\
R_{p,t}^{i,j} &= \sum_{l \in \mathcal{S}(i,j)} \frac{w_{l,t}^{i,j}}{\sum_{l \in \mathcal{S}(i,j)} w_{l,t}^{i,j}} R_{l,t}, \\
\mathcal{R}_{p,j-,t}^i &= \sum_{l \neq j, l=1}^{J(i)} \frac{A_t^{i,l}}{\sum_{l \neq j, l \in J(i)} A_t^{i,l}} R_{p,t}^{i,l},
\end{aligned} \tag{47}$$

where $w_{l,t}^{i,j}$ is mutual fund j 's allocation to country l , $A_t^{i,l}$ is fund l total assets under management and $R_{l,t}$ is country l 's equity MSCI return in month t . We then compute the covariances by country and fund pair, and compute $\Delta Cov_k^{i,j}$ as the differential

$$\Delta Cov_k^{i,j} = \frac{Cov(R_k, R_{p,k-}^{i,j})}{Cov(R_k, R_k - R_{p,k-}^{i,j})} - \frac{Cov(R_k, \mathcal{R}_{p,j-}^{i,j})}{Cov(R_k, R_k - R_{p,k-}^{i,j})}$$

B.2 Summary Statistics for $\Delta Cov^{i,j}$

Table B.1: Summary Statistics for $\Delta Cov_k^{i,j}$

Variable	Mean	Median	S.D.	p25	p75
$\Delta Cov^{i,j}$	.05	-.008	.51	-.13	.14

B.3 Imputation of Expectations

We assume that, for each country, expectations can be decomposed into a common term and an investor-specific term:

$$E_t^i(g_k^{\text{next year}}) = \bar{E}_t(g_k^{\text{next year}}) + U_{t,k}^i \tag{48}$$

We make the identifying assumption that $E(U_{t,k}^i) = 0$, so that $\bar{E}_t(g_k^{\text{next year}})$ can be estimated as the cross-investor average $\bar{E}_t(g_k^{\text{next year}}) = \frac{1}{M} \sum_{i=1}^M E_t^i(g_k^{\text{next year}})$, and $U_{t,k}^i$ as the residual: $U_{t,k}^i = E_t^i(g_k^{\text{next year}}) - \frac{1}{M} \sum_{i=1}^M E_t^i(g_k^{\text{next year}})$.

We assume that the investor-specific term $U_{t,k}^i$ is the sum of a year-specific term and a

month-specific term that are independent from each other:

$$U_{t,k}^i = \bar{U}_{year,k}^i + u_{year,month,k}^i \quad (49)$$

where $t = 12 \times year + month$. We make the identifying assumption that $E(u_{year,month,k}^i) = 0$, so that $\bar{U}_{year,k}^i$ can be estimated as $\bar{U}_{year,k}^i = \frac{1}{12} \sum_{month=1}^{12} U_{year \times 12 + month,k}^i$, and $u_{year,month,k}^i = U_{year \times 12 + month,k}^i - \frac{1}{12} \sum_{month=1}^{12} U_{year \times 12 + month,k}^i$. By construction, $E(\bar{U}_{year,k}^i) = 0$.

We allow the year-specific component $\bar{U}_{year,k}^i$ to be autocorrelated:

$$\bar{U}_{year,k}^i = \rho^v \bar{U}_{year-1,k}^i + v_{k,year}^i \quad (50)$$

with $v_{k,year}^i \sim N(0, \sigma_k^v)$. The autocorrelation parameter ρ^v is common across countries and investors, but the variance of the innovation σ_k^v is country-specific.

We fit the autoregressive process (50) on $\bar{U}_{year,k}^i$ to estimate ρ^v . The country-specific standard deviation σ_k^v is estimated as the standard deviation of the residuals of the autoregressive equation.

We allow the month-specific component $u_{year,month,k}^i$ to be autocorrelated:

$$u_{year,month,k}^i = \rho^e u_{year,month-1,k}^i + e_{year,month,k}^i \quad (51)$$

with $e_{year,month,k}^i \sim N(0, \sigma_k^e)$. The autocorrelation parameter ρ^e is common across countries and investors, but the variance of the innovation σ_k^e is country-specific.

We fit the autoregressive process (51) on $u_{year,month,k}^i$ to estimate ρ^e . The country-specific standard deviation σ_k^e is estimated as the standard deviation of the residuals of the autoregressive equation.

These estimations are performed on the subset of investors and countries for which we have expectation data. We then impute expectations for the investors with missing country expectations as follows:

$$\widehat{E}_t^i(g_k^{\text{next year}}) = \bar{E}_t(\widehat{g_k^{\text{next year}}}) + \widehat{\bar{U}_{year,k}^i} + \widehat{u_{year,month,k}^i} \quad (52)$$

where $\bar{E}_t(\widehat{g_k^{\text{next year}}})$ is the estimated consensus, and $\widehat{\bar{U}_{year,k}^i}$ and $\widehat{u_{year,month,k}^i}$ are simulated using the data-generating processes (50) and (51), using our estimates of ρ^v , ρ^e , σ_k^v and σ_k^e .

C IV Appendix

This appendix formalizes the argument behind our use of the IMF growth forecast as an instrumental variable for the in-house Consensus forecast. We work in a stylized environment in which the forecaster who produces the in-house forecast and the decision-maker (the fund manager or the end-investor) form their forecasts as linear functions of a common public signal (the IMF forecast) and of distinct private signals. We do *not* impose Bayesian rationality on either agent, and we allow the noise of the two private signals to have different variances. We show that, as long as the two agents weigh signals in the same way, the OLS estimator that uses the in-house forecast as a proxy for the decision-maker’s forecast is attenuated by a measurement-error bias, while the IV estimator that uses the IMF forecast as an instrument recovers the structural elasticity β . When the two agents weight signals differently, the IV is biased, but a complementary pair of estimators that augment the instruments with controls (the IMF-forecast IV controlling for vintage growth, and the vintage-growth IV controlling for the IMF forecast) biases β in opposite directions and brackets β under standard behavioral models of expectation formation.

C.1 Setup

Let g denote the next-year GDP growth of a given country at a given date; throughout this appendix we suppress the country and time indices. The prior on g is $g \sim \mathcal{N}(0, \sigma_g^2)$, where the zero mean is without loss of generality. Three pieces of information about g are available: a public signal embodied by the IMF forecast,

$$s^P = g + u^P, \quad u^P \sim \mathcal{N}(0, \sigma_P^2),$$

a private signal of the forecaster $s^F = g + u^F$, with $u^F \sim \mathcal{N}(0, \sigma_F^2)$, and a private signal of the decision-maker $s^D = g + u^D$, with $u^D \sim \mathcal{N}(0, \sigma_D^2)$. The shocks g , u^P , u^F and u^D are mutually independent.²⁴ Rather than imposing Bayes’ rule, we allow each agent to combine the signals into a forecast in an arbitrary linear way:

$$E^F[g] = a_P^F s^P + a^F s^F, \tag{53}$$

$$E^D[g] = a_P^D s^P + a^D s^D, \tag{54}$$

²⁴We treat the IMF forecast as the only public signal. The argument extends to the case in which s^P is itself a function of the IMF forecast and other public signals.

where any nonzero intercepts are absorbed by the fixed effects in the empirical specification. The structural equation that the empirical model maps to is

$$y = \beta E^D[g] + \varepsilon, \quad (55)$$

where y is the (log) allocation and ε collects all the determinants of the allocation that are orthogonal to expectations and absorbed by the fixed effects of Equation (17). As a baseline, we assume that the forecaster and the decision-maker weigh signals in the same way:

$$(a_P^F, a^F) = (a_P^D, a^D) \equiv (a_P, a). \quad (56)$$

We relax this assumption in Section C.3.

C.2 Measurement-error bias and the IMF instrument

Under the symmetry restriction (56), the wedge between the in-house forecast and the decision-maker's forecast is

$$\xi \equiv E^F[g] - E^D[g] = a(u^F - u^D),$$

with mean zero and variance $\text{Var}(\xi) = a^2(\sigma_F^2 + \sigma_D^2)$. The wedge is correlated with the true regressor, $\text{Cov}(E^D[g], \xi) = -a^2\sigma_D^2$. The measurement error is non-classical, in the sense that ξ is not orthogonal to $E^D[g]$, but it is orthogonal to the public signal,

$$\text{Cov}(s^P, \xi) = 0, \quad (57)$$

because u^P is independent of (u^F, u^D) . The two forecasts have the following unconditional variances

$$V_F \equiv \text{Var}(E^F[g]) = C + a^2\sigma_F^2, \quad V_D \equiv \text{Var}(E^D[g]) = C + a^2\sigma_D^2, \quad (58)$$

where the common term depends on the variance of the public signal and of the fundamental:

$$C \equiv \text{Cov}(E^F[g], E^D[g]) = a_P^2(\sigma_g^2 + \sigma_P^2) + 2a_P a \sigma_g^2 + a^2\sigma_g^2.$$

OLS bias. Replacing $E^D[g]$ with the observable in-house forecast $E^F[g]$ in (55), the OLS estimator converges to

$$\text{plim } \hat{\beta}_{OLS} = \beta \frac{\text{Cov}(E^F[g], E^D[g])}{\text{Var}(E^F[g])} = \beta \frac{C}{C + a^2 \sigma_F^2} = \beta \left(1 - \frac{a^2 \sigma_F^2}{V_F}\right) < \beta. \quad (59)$$

OLS attenuates the structural elasticity. The size of the attenuation is governed by the share of the variance of the in-house forecast that is driven by the forecaster's private signal: when σ_F^2 is large relative to the rest of the variation in $E^F[g]$, the in-house forecast is a noisy proxy for the decision-maker's forecast and OLS is severely attenuated.

IV with the IMF forecast. The public signal s^P is relevant for the in-house forecast,

$$\text{Cov}(s^P, E^F[g]) = a_P(\sigma_g^2 + \sigma_P^2) + a \sigma_g^2 > 0, \quad (60)$$

and satisfies the exclusion restriction $\text{Cov}(s^P, \xi) = 0$ by (57) and $\text{Cov}(s^P, \varepsilon) = 0$ by assumption. The IV estimator then converges to

$$\text{plim } \hat{\beta}_{IV} = \frac{\text{Cov}(s^P, y)}{\text{Cov}(s^P, E^F[g])} = \beta \frac{\text{Cov}(s^P, E^D[g])}{\text{Cov}(s^P, E^F[g])} = \beta, \quad (61)$$

where the last equality uses $\text{Cov}(s^P, E^F[g]) = \text{Cov}(s^P, E^D[g]) = a_P(\sigma_g^2 + \sigma_P^2) + a \sigma_g^2$. The key step is that these two covariances depend only on the weights (a_P, a) and the public-signal precision σ_P^2 . They do not depend on σ_F^2 or σ_D^2 . Asymmetry in the precision of the two private signals therefore does not contaminate the IV: the public signal s^P traces out variation in $E^F[g]$ that is identical to the variation it would induce in $E^D[g]$, and the IV recovers β exactly.

C.3 Asymmetric weights and bounds analysis

The symmetry of the weights in (56) is the key identifying assumption. If $(a_P^F, a^F) \neq (a_P^D, a^D)$, the IMF-forecast IV is no longer consistent and converges to

$$\text{plim } \hat{\beta}_{IV}^{\text{IMF}} = \beta \frac{a_P^D(\sigma_g^2 + \sigma_P^2) + a^D \sigma_g^2}{a_P^F(\sigma_g^2 + \sigma_P^2) + a^F \sigma_g^2} = \beta \frac{a_P^D r + a^D}{a_P^F r + a^F}, \quad (62)$$

where $r \equiv (\sigma_g^2 + \sigma_P^2)/\sigma_g^2 > 1$ is the ratio of the variance of the public signal to the variance of the fundamental. To bound the structural elasticity β , we consider two additional estimators that augment the instrument with a control variable: (i) the IMF-forecast IV with realized growth g added as a control, denoted $\hat{\beta}_{IV}^{\text{IMF}|g}$; and (ii) the vintage-growth IV with the IMF

forecast s^P added as a control, denoted $\hat{\beta}_{IV}^{g|IMF}$. Under the general specification (53)-(54), these estimators converge to

$$\text{plim } \hat{\beta}_{IV}^{IMF|g} = \beta \frac{a_P^D}{a_P^F}, \quad (63)$$

$$\text{plim } \hat{\beta}_{IV}^{g|IMF} = \beta \frac{a^D}{a^F}. \quad (64)$$

The two controlled estimators isolate the two “primitive” loading ratios. The IMF-forecast IV with vintage-growth control uses, by Frisch-Waugh, the residual instrument $\tilde{s}^P = s^P - g = u^P$, the pure IMF noise. Because u^P is uncorrelated with g , s^F and s^D , only the public-signal loadings (a_P^F, a_P^D) enter the bias formula. Symmetrically, the vintage-growth IV with IMF-control uses the residual instrument $\tilde{g} = g - \lambda s^P$, with $\lambda = \sigma_g^2 / (\sigma_g^2 + \sigma_P^2)$, which orthogonalizes the fundamental against the public-signal variation; only the private-signal loadings (a^F, a^D) enter the bias formula.²⁵ We now show that under three natural specifications of expectation formation: heterogeneous subjective (or objective) signal precisions, heterogeneous strategic concerns, and nested information sets, the two controlled estimators bracket the structural elasticity β .

Heterogeneous subjective precisions Assume that each agent forms their forecast as a Bayesian posterior mean given a set of *subjective* precisions for the prior, the public signal, and their own private signal: $a_P^i = \tilde{\tau}_P / T^i$ and $a^i = \tilde{\tau}^i / T^i$ with $T^i \equiv \tilde{\tau}_g + \tilde{\tau}_P + \tilde{\tau}^i$, for $i \in \{F, D\}$. The subjective precisions $\tilde{\tau}_g$, $\tilde{\tau}_P$ and $\tilde{\tau}^i$ need not equal their objective counterparts, so the weights are not necessarily rational. This formulation relates to the behavioral-finance literature on overconfidence in own private information (e.g., Daniel et al., 1998) and to the macroeconomic-expectations literature documenting heterogeneous over- and under-reactions in survey forecasts (e.g., Bordalo et al., 2020; Broer and Kohlhas, 2024; Adam et al., 2025). We assume that the two agents share the same subjective prior and public-signal precisions, and that the only source of asymmetry between them is their subjective perception of the precision of their own private signal: $\tilde{\tau}_F \neq \tilde{\tau}_D$. Suppose the decision-maker perceives their private signal as more precise than the forecaster does, $\tilde{\tau}_D > \tilde{\tau}_F$. Direct computation gives

$$\frac{a_P^D}{a_P^F} = \frac{T^F}{T^D} < 1 \quad \text{and} \quad \frac{a^D}{a^F} = \frac{\tilde{\tau}_D}{\tilde{\tau}_F} \cdot \frac{T^F}{T^D} > 1. \quad (65)$$

²⁵Both controls require the exogeneity restrictions $\text{Cov}(g, \varepsilon) = 0$ and $\text{Cov}(s^P, \varepsilon) = 0$, which are the same restrictions that support the use of g and s^P as instruments. The IMF-forecast IV without control, (62), lies between these two pure ratios by the median inequality: it interpolates between a_P^D/a_P^F and a^D/a^F as a function of r .

The IMF-forecast IV with vintage-growth control therefore biases β *downward*, and the vintage-growth IV with IMF-control biases β *upward*. The two estimators bracket the structural elasticity:

$$\text{plim } \hat{\beta}_{IV}^{\text{IMF}|g} \leq \beta \leq \text{plim } \hat{\beta}_{IV}^{g|\text{IMF}}. \quad (66)$$

The bounds coincide with β when the symmetry restriction (56) holds. Symmetrically, when $\tilde{\tau}_F > \tilde{\tau}_D$ (the forecaster has the more aggressive private-signal perception), the inequalities in (66) reverse, and the two estimators bracket β from the opposite direction.

Heterogeneous strategic concerns A second potential source of asymmetric weights arises when the forecaster faces career or reputational incentives that the decision-maker does not share. Forecasting teams are routinely evaluated against the cross-section of professional forecasters, so deviating too far from the public benchmark is professionally costly; the decision-maker, by contrast, is evaluated on portfolio performance and has no analogous incentive to anchor on the IMF forecast. Following the strategic-forecasting literature (e.g., Ottaviani and Sørensen, 2006), we capture this asymmetry by letting the forecaster shift their forecast partially toward the public signal,

$$E^F[g] = (1 - \mu) E^*[g | s^P, s^F] + \mu s^P, \quad \mu \in [0, 1), \quad (67)$$

where $E^*[g | s^P, s^F] \equiv w_P s^P + w s^F$ is the rational Bayesian forecast and μ parameterizes the strategic anchoring intensity. The decision-maker remains free of strategic distortion, $E^D[g] = w_P s^P + w s^D$. The implied weights are

$$a_P^F = w_P + \mu(1 - w_P), \quad a^F = (1 - \mu)w, \quad a_P^D = w_P, \quad a^D = w, \quad (68)$$

so strategic anchoring raises the forecaster's loading on the public signal and lowers their loading on the private signal: $a_P^F > a_P^D$ and $a^F < a^D$. Direct computation gives

$$\frac{a_P^D}{a_P^F} = \frac{w_P}{w_P + \mu(1 - w_P)} < 1 \quad \text{and} \quad \frac{a^D}{a^F} = \frac{1}{1 - \mu} > 1, \quad (69)$$

so the same bracketing result obtains as in the heterogeneous-precision case:

$$\text{plim } \hat{\beta}_{IV}^{\text{IMF}|g} \leq \beta \leq \text{plim } \hat{\beta}_{IV}^{g|\text{IMF}}. \quad (70)$$

Under both behavioral models, the IMF-forecast IV with vintage-growth control and the vintage-growth IV with IMF-control bound the structural elasticity β from below and above. Symmetrically, when the decision-maker strategically aligns on public information, while the

forecaster does not, the inequalities in (70) reverse, and the two estimators bracket β from the opposite direction. The results carry through in the more general case where strategic motives are present for both, but with different strength.²⁶

Nested information A third case of interest is when the decision-maker does not form her expectation from scratch but observes the in-house forecast $E^F[g]$ and combines it with her own private signal s^D . We discipline her weights through subjective Bayesian updating with precisions *shared* with the forecaster: both agents perceive the prior with precision $\tilde{\tau}_g$, the public signal with precision $\tilde{\tau}_P$, the forecaster's private signal with precision $\tilde{\tau}_F$, and the decision-maker's own private signal with precision $\tilde{\tau}_D$. The forecaster's weights are then as in the heterogeneous-precision paragraph,

$$a_P^F = \frac{\tilde{\tau}_P}{T^F}, \quad a^F = \frac{\tilde{\tau}_F}{T^F}, \quad T^F \equiv \tilde{\tau}_g + \tilde{\tau}_P + \tilde{\tau}_F,$$

and the decision-maker, knowing that $E^F[g]$ summarizes (s^P, s^F) through this rule, performs a Bayesian update of g on (s^P, s^F, s^D) , equivalently, on (E^F, s^D) . The resulting forecast is

$$E^D[g] = \alpha E^F[g] + \gamma s^D, \quad \alpha = \frac{T^F}{T^D}, \quad \gamma = \frac{\tilde{\tau}_D}{T^D}, \quad T^D \equiv T^F + \tilde{\tau}_D, \quad (71)$$

with $\alpha + \gamma \leq 1$. Because $E^F[g]$ already incorporates the shared prior and the public signal, the decision-maker does not double-count them, and the weights α and γ are pinned down by the same primitive precisions that govern a^F . Substituting (53), the decision-maker's expectation loads on three signals: the public signal, the forecaster's private signal s^F (through E^F), and her own private signal s^D , with weights $\tilde{\tau}_P/T^D$, $\tilde{\tau}_F/T^D$ and $\tilde{\tau}_D/T^D$ respectively, which is precisely the Bayesian posterior on the full information set. Substituting into (63) and (64) yields

$$\text{plim } \hat{\beta}_{IV}^{\text{IMF}|g} = \beta \alpha = \beta \frac{T^F}{T^D}, \quad (72)$$

$$\text{plim } \hat{\beta}_{IV}^{g|\text{IMF}} = \beta \left(\alpha + \frac{\gamma}{a^F} \right) = \beta \frac{T^F}{T^D} \cdot \frac{\tilde{\tau}_F + \tilde{\tau}_D}{\tilde{\tau}_F}. \quad (73)$$

²⁶Gemmi and Valchev (2025) argue that professional forecasters in fact *diversify* from consensus (overweighting their private signal and underweighting the public signal) the opposite direction from (67). This corresponds to $E^F[g] = (1 - \nu) E^*[g | s^P, s^F] + \nu s^F$ with $\nu \in [0, 1)$, which delivers $a_P^F = (1 - \nu)w_P < a_P^D$ and $a^F = w + \nu(1 - w) > a^D$; under this alternative the inequalities in (70) reverse, with the vintage-growth IV with IMF-control becoming the lower bound and the IMF-forecast IV with vintage-growth control becoming the upper bound, but the bracketing of β continues to hold.

The IMF-instrument estimator understates β because $T^F/T^D \leq 1$, and the vintage-instrument estimator overstates β because $(T^F/T^D)(\tilde{\tau}_F + \tilde{\tau}_D)/\tilde{\tau}_F \geq 1$. The latter follows directly from $T^F(\tilde{\tau}_F + \tilde{\tau}_D) - T^D\tilde{\tau}_F = (\tilde{\tau}_g + \tilde{\tau}_p)\tilde{\tau}_D \geq 0$. The bracket

$$\text{plim } \hat{\beta}_{IV}^{\text{IMF}|g} \leq \beta \leq \text{plim } \hat{\beta}_{IV}^{g|\text{IMF}} \quad (74)$$

therefore holds in the nested information case.

D Additional Tables

Table D.1: Equity returns and GDP growth

	(1)	(2)	(3)	(4)
	$\Delta \log(MSCI_t)$	$\Delta \log(MSCI_t)$	$\Delta \log(MSCI_t)$	$\Delta \log(MSCI_t)$
VARIABLES				
$g_{k,t}$	2.122*** (0.218)	2.557*** (0.260)	2.511*** (0.257)	2.550*** (0.248)
dlmsci1			0.152*** (0.033)	0.041 (0.034)
$\Delta \log(MSCI_{t-1})$				2.231*** (0.255)
Observations	792	792	764	743
R-squared	0.107	0.114	0.140	0.216
Country FE	No	Yes	Yes	Yes

Notes: Driscoll–Kraay standard errors in parentheses. The table reports results from annual panel regressions of equity returns on contemporaneous GDP growth. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.2: Sample build-down

Sample	Obs.	Countries	Funds	Investors	Median AUM
EPFR allocations (full)	4,416,216	51	3,346	604	338.1
Matched sample					
+ matched with subj. exp.	1,308,600	49	1,059	54	333.7
OLS sample					
+ clean with subj. exp.	168,190	46	827	50	351.4
+ fund-level subj. exp. and controls	38,654	37	213	13	269.7
IV sample					
+ clean with IMF exp.	541,961	49	1,055	54	318.6
+ fund-level IMF exp. and controls	68,837	40	790	48	298.1
Quantitative sample					
+ clean with imputed exp.	1,285,308	45	2,921	549	361.1

Notes: The table reports the size of each sample used in the empirical and quantitative analysis. The full EPFR sample includes all country–fund–month observations with at least one strictly positive allocation. The matched sample corresponds to the EPFR funds with at least one matched expectation from Consensus Economics. The OLS sample restricts to country–fund pairs with mean allocation exceeding 0.5%, matched with Consensus Economics GDP forecasts. The IV sample applies the same allocation threshold but uses IMF World Economic Outlook forecasts. The fund-level rows additionally require fund-level expectations and the controls (covariance with equity returns and lagged MSCI returns) used in the regressions. The quantitative sample corresponds to the EPFR sample matched with imputed expectations and is restricted to fund–month observations with positions in more than five countries. Median AUM is the median of fund-level assets under management across funds, in USD millions.

Table D.3: Number of observations per institutions

Institution	Matched	Clean with subj. exp.	+ fund-level & controls
JPMORGAN ASSET MANAGEMENT	82,889	38,665	12,790
UBS GLOBAL ASSET MANAGEMENT	43,533	28,258	10,663
DWS INVESTMENTS	33,038	16,963	2,703
MORGAN STANLEY INVESTMENT MANAGEMENT	24,484	14,966	298
DEUTSCHE ASSET MANAGEMENT	20,971	11,929	795
HSBC GLOBAL ASSET MANAGEMENT	18,431	12,735	5,877
GOLDMAN SACHS ASSET MANAGEMENT	12,162	9,295	3,960
CREDIT SUISSE ASSET MANAGEMENT	9,393	6,711	651
ALLIANZ GLOBAL INVESTORS	8,961	6,878	53
SCHRODER INVESTMENT MANAGEMENT	8,497	5,404	0
BNP PARIBAS INVESTMENT PARTNERS	4,450	3,105	450
ING INVESTMENT MANAGEMENT	2,548	1,320	184
ZURCHER KANTONALBANK	1,578	1,132	0
DEKA INVESTMENT GMBH	1,563	1,464	0
AMUNDI ASSET MANAGEMENT	1,479	1,108	0
SOCIETE GENERALE ASSET MANAGEMENT	1,221	1,172	0
NOMURA ASSET MANAGEMENT	1,210	509	228
HANDELSBANKEN ASSET MANAGEMENT	1,136	650	0
RAIFFEISEN CAPITAL MANAGEMENT	948	557	0
BNP PARIBAS ASSET MANAGEMENT	797	758	0
PICTET ASSET MANAGEMENT	708	652	0
DNB ASSET MANAGEMENT AS	573	502	0
BMO ASSET MANAGEMENT	559	201	0
ABN AMRO ASSET MANAGEMENT	513	471	2
FIRST TRUST ADVISORS	500	304	0
AXA INVESTMENT MANAGERS	479	453	0
JULIUS BAER	396	384	0
DWS GROUP	389	283	0
INVESCO ASSET MANAGEMENT	239	209	0
ITAU ASSET MANAGEMENT	168	142	0
VONTOBEL ASSET MANAGEMENT	159	146	0
DB PLATINUM ADVISORS	149	79	0
NORDEA INVESTMENT MANAGEMENT	148	60	0
BANCO SANTANDER CENTRAL HISPANO MADRID	139	114	0
COMMERZBANK AG	132	132	0
LA BANQUE POSTALE ASSET MANAGEMENT	99	99	0
WELLS CAPITAL MANAGEMENT	73	0	0
LBBW ASSET MANAGEMENT	72	72	0
SEB INVESTMENT MANAGEMENT	64	43	0
NATIXIS ASSET MANAGEMENT	55	55	0
ERSTE-SPARINVEST	52	51	0
NORTHERN TRUST INVESTMENTS	52	1	0
DANSKE CAPITAL	50	0	0
Others	202	158	0

Notes: The table reports the number of country–fund–month observations by institution across three nested samples. Matched: observations with Consensus Economics GDP forecasts available. Clean with subj. exp.: adds mean allocation exceeding 0.5% and non-missing log allocation. + fund-level & controls: additionally requires fund-level granular subjective expectations, the covariance control, and lagged MSCI returns. Institutions are sorted in descending order of matched-sample observations. Institutions with fewer than 50 matched-sample observations are aggregated into the row.

Table D.4: Fund-Country Allocations and Investor Expectations – Passive and Active Funds

	(1)	(2)	(3)	(4)
	$\log(A_{k,t}^{i,j})$	$\log(A_{k,t}^{i,j})$	$\log(A_{k,t}^{i,j})$	$\log(A_{k,t}^{i,j})$
	Passive	Active	Passive	Active
			IV (boot SE)	IV (boot SE)
VARIABLES				
$E_t^i(g_k^{\text{next year}})$	0.013	0.028***	-0.017	0.067***
	(0.013)	(0.009)	(0.011)	(0.014)
$\Delta \log(Q_{k,t})$			0.451***	0.615***
			(0.033)	(0.025)
$\Delta \log(Q_{k,t-1})$			0.429***	0.590***
			(0.032)	(0.025)
Observations	36,483	112,507	115,489	371,403
R-squared	0.991	0.968	0.979	0.926
Country-fund FE	Yes	Yes	Yes	Yes
Country-time FE	Yes	Yes	No	No
Fund-time FE	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses. Columns (1)–(2) are OLS with analytic standard errors clustered at the investor–country level. Columns (3)–(4) report two-step IV estimates that use generated regressors from first-stage projections; their standard errors are computed via a pairs cluster bootstrap with 200 replications and clusters at the investor–country level, which propagates the first-stage estimation uncertainty into the reported standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

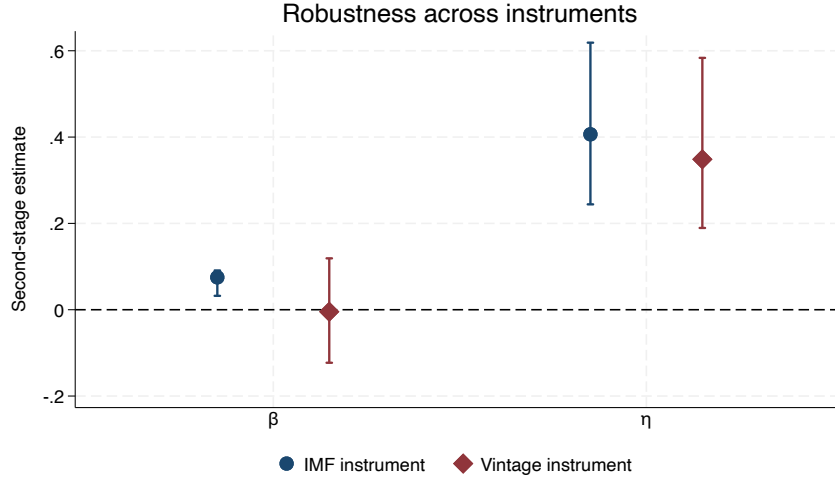
Table D.5: Robustness - Contribution of co-ownership spillovers

Specification	Avg. co-ownership share [p10, p90]
Baseline ($p = 0.13$)	60% [25%, 85%]
Conservative ($p = 0.18$)	46% [14%, 74%]
Liberal ($p = 0.07$)	83% [60%, 96%]
Constant weights	58% [19%, 87%]
Min. 6 obs. per country-date	57% [11%, 88%]
Consensus expectations	62% [26%, 87%]

Notes: The table reports the adjusted contribution of co-ownership spillovers to the cross-country variance of expectation-driven capital-flow reallocation – i.e., $V(\eta(\Gamma_{k,t} - \Gamma_{k,k,t}))$ as a share of the variance of $\tilde{\Delta}a_{k,t}$ – under alternative calibrations of the portfolio adjustment probability $p = \beta/(\beta + \eta)$ and under alternative measurement choices. Reported values are cross-country averages; brackets show the $[p_{10}, p_{90}]$ range. *Baseline* uses $p = 0.13$ and the actual fund-level portfolio weights $w_{k,t}^{i,j}$. *Conservative* uses the upper-bound estimate of $p = 0.18$ from Section 3.3. *Liberal* uses the lower estimate of $p = 0.07$ taken from Bacchetta and van Wincoop (2017). *Constant weights* fixes the fund weights $w_{k,t}^{i,j}$ at their sample averages $\bar{w}_k^{i,j}$. *Min. 6 obs. per country-date* imposes a minimum of 6 observations to compute average forecasts in our imputation procedure. *Consensus expectations* replaces the investor-specific (partially imputed) expectations $l_{k,t}^i$ with the cross-investor consensus $l_{k,t} = (\sum_i l_{k,t}^i)/M$.

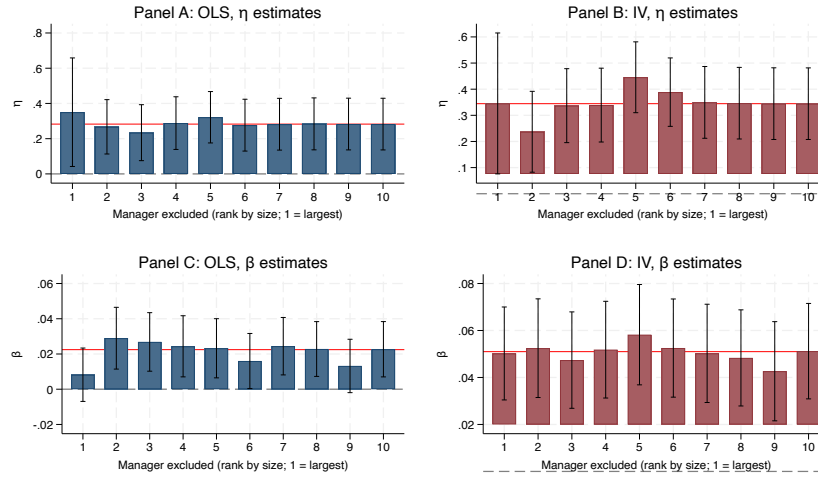
E Additional Figures

Figure E.1: Robustness – IMF forecasts vs vintage growth instruments



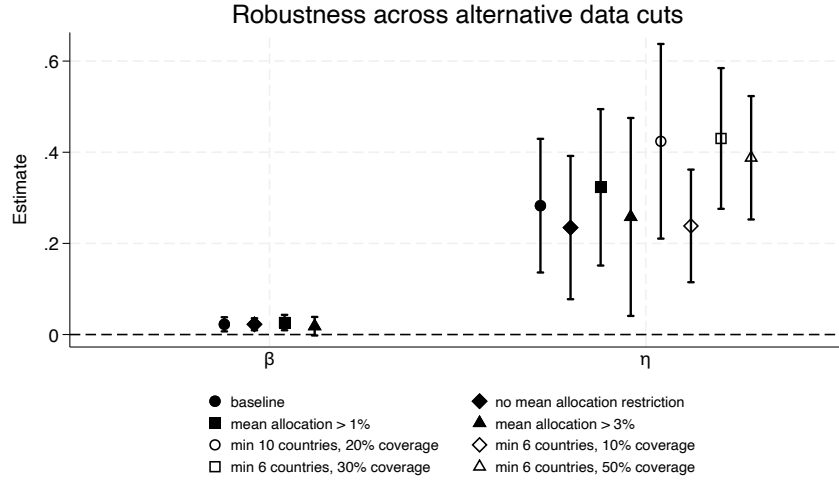
Notes: The figure reports IV estimates of β and η coefficient under two complementary specifications: the IMF-forecast IV controlling for realized (vintage) GDP growth, and the vintage-growth IV controlling for the IMF forecast. Markers are point estimates and whiskers are 95% confidence intervals based on bootstrapped standard errors clustered on investor and country.

Figure E.2: Robustness – Leave-one-out estimates



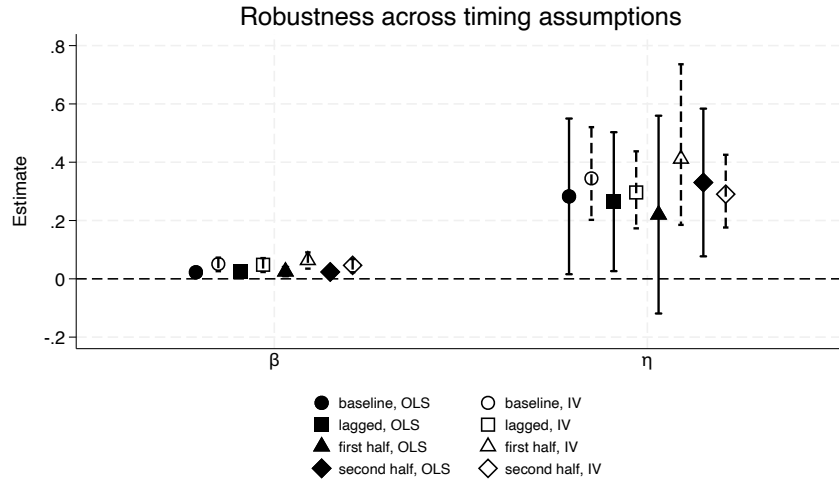
Notes: The figure reports leave-one-out estimates of β and η obtained by re-estimating the OLS and IV specifications after dropping, one at a time, each of the ten managers with the largest number of observations in the baseline sample. Bars are point estimates and whiskers are 95% confidence intervals based on standard errors clustered on investor and country. Managers are ranked by their number of observations in the baseline sample, with rank 1 the largest. The horizontal dashed line in each panel marks the full-sample baseline estimate. Fixed-effect structure, instruments, and clustering follow the corresponding baseline specification.

Figure E.3: Robustness – Alternative data cuts



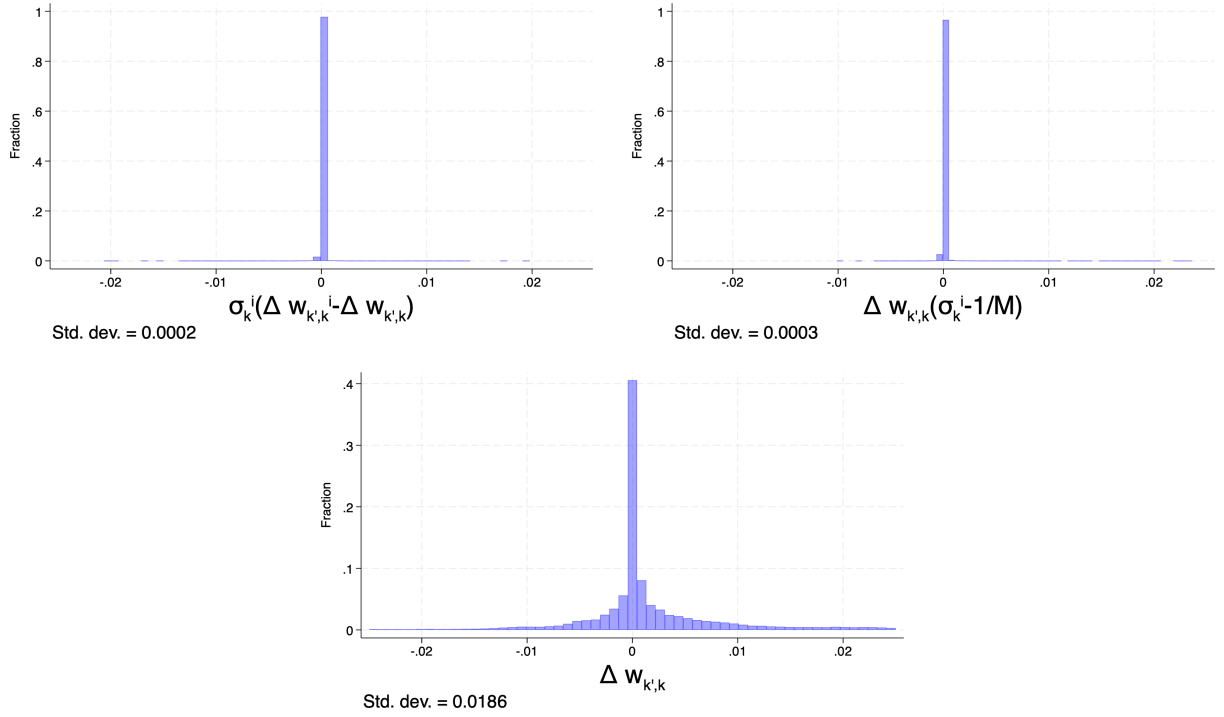
Notes: The figure reports the OLS estimates of β and η coefficient under alternative sample restrictions. Markers are point estimates and whiskers are 95% confidence intervals based on standard errors clustered on investor and country. In the baseline, the sample is restricted to country-fund pairs with a mean fund allocation of 0.5% and the fund expectations are computed only if the expectation data covers at least 20% of the fund portfolio and a minimum of 6 countries. The leftmost marker in each panel reproduces the baseline estimate.

Figure E.4: Robustness – Timing and horizon



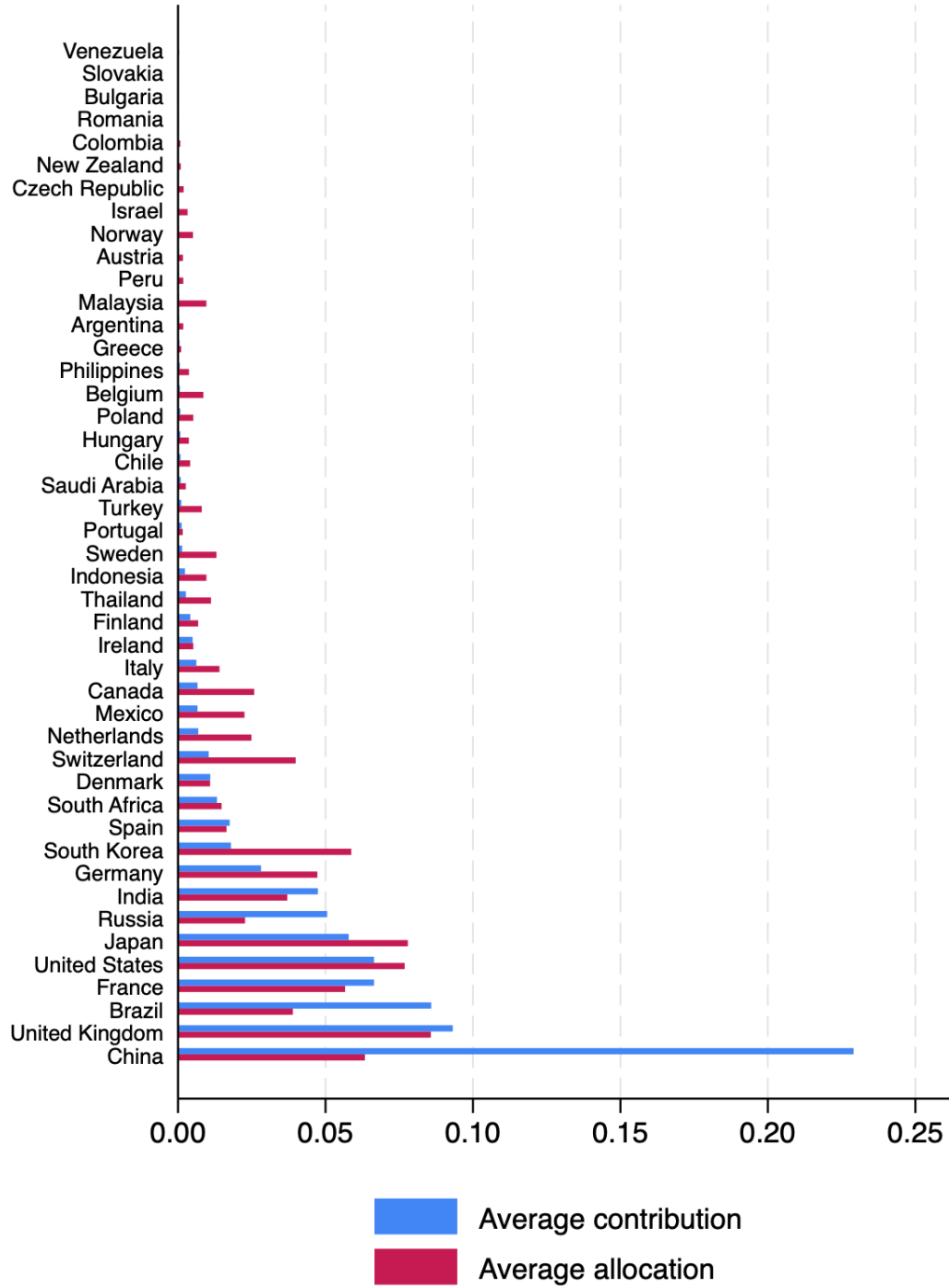
Notes: The figure reports the OLS estimates of β and η coefficient under alternative timing and horizon specifications. “Lagged” uses forecasts and instruments observed one month before allocation; “first half” / “second half” restrict the sample to months 1–6 / 7–12 of the calendar year. Markers are point estimates and whiskers are 95% confidence intervals based on standard errors clustered on investor and country. Filled markers are OLS, hollow markers are IV instrumented with the IMF forecast.

Figure E.5: Distribution of weights



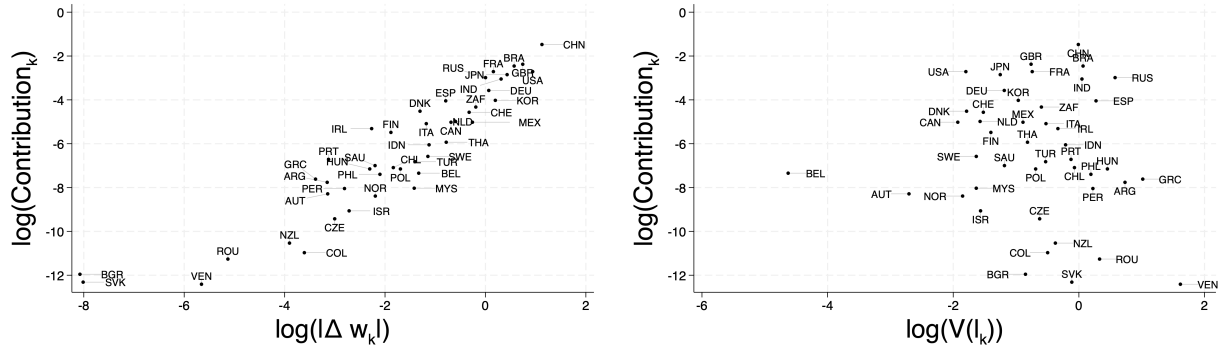
Notes: The left panel represents the distribution and standard deviation of $\sigma_{k,t}^i(\Delta w_{k,k',t}^i - \Delta w_{k,k',t})$ across country pairs and investors. The right panel represents the distribution and standard deviation of $\Delta w_{k,k',t}(\sigma_{k,t}^i - 1/M)$ across country pairs and investors. The bottom panel represents the distribution and standard deviation of $\Delta w_{k,k',t}$ across country pairs.

Figure E.6: Contributors to co-ownership spillovers



Notes: The figure represents the scatter plot of the average weights $w_k = \sum_{i=1}^M \sigma_k^i w_k^i$ against the average contributions $Contribution_k = \sum_{k'=1}^K \sigma_{k'} Contribution_{k',k}$.

Figure E.7: Role of weights and idiosyncratic volatility



Notes: The left panel represents the scatter plot of the log of the average absolute value of co-ownership linkages $|\Delta w_k| = \sum_{k'=1}^K \sigma_{k'} |\Delta w_{k',k}|$ against the log of average contributions $Contribution_k = \sum_{k'=1}^K \sigma_{k'} Contribution_{k',k}$. The right panel represents the scatter plot of the log of the variance of country-specific expectations $V(l_k)$ against the log of average contributions $Contribution_k = \sum_{k'=1}^K \sigma_{k'} Contribution_{k',k}$.